AnovaCraft Quick Guide

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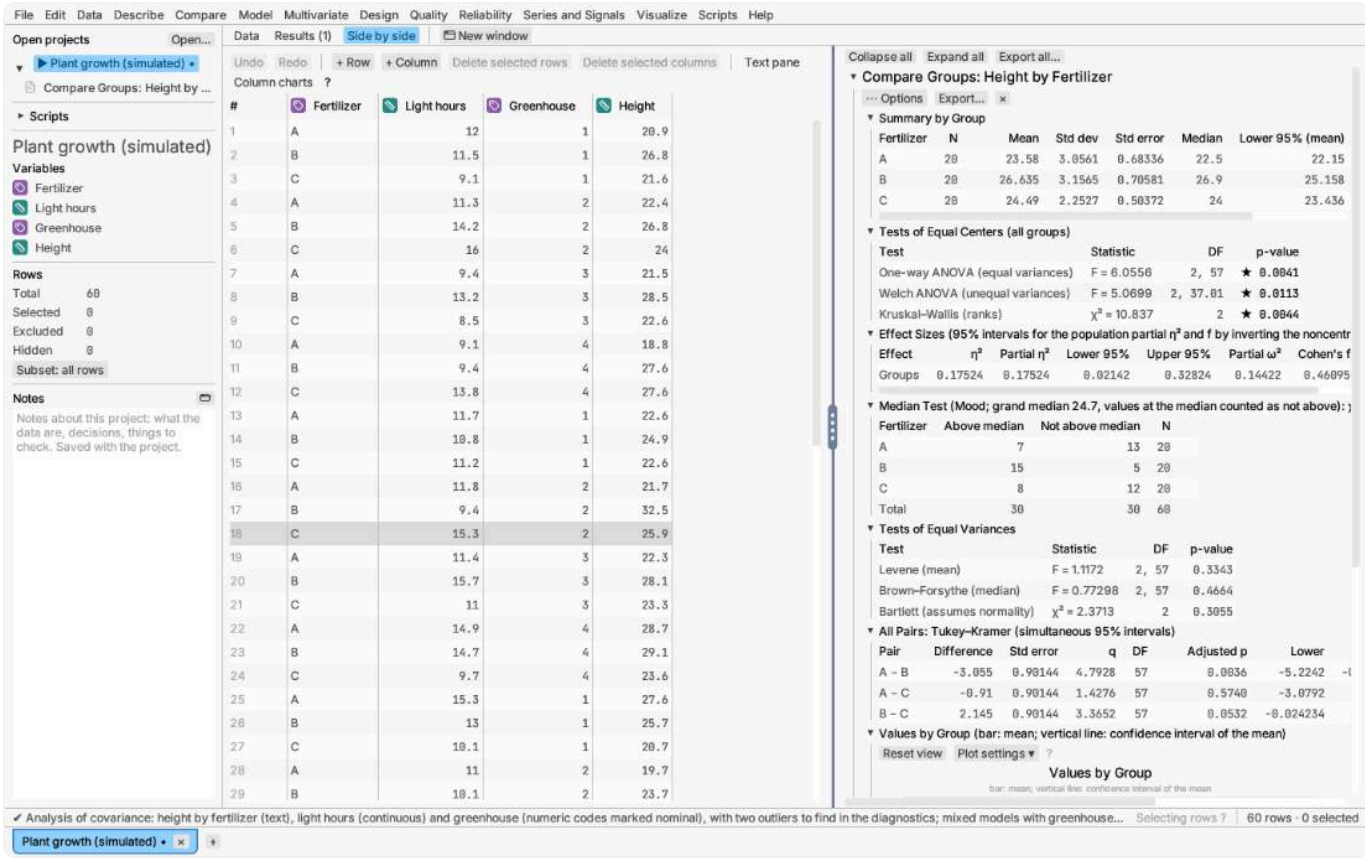
Start here

AnovaCraft is a free program for looking at data and answering questions with it: what is typical, whether groups differ, what drives a result, how to plan an experiment, and whether a process is behaving.

This Quick Guide is the short version. It explains, in plain words, what every menu item does and when you would use it. When you want the details (every option, the method behind a result, how to script it), each entry links to the **Reference Manual**.

How the app works, in one minute

1. **Open some data.** A table of rows and columns, like a spreadsheet. Each row is one thing you measured (a person, a part, a day) and each column is one measurement. File ▸ Open sample data has dozens of ready-made tables to practise on.
2. **Pick an analysis from a menu.** The menus are named for the question you are asking: Describe, Compare, Model and so on. A dialog asks which columns to use.
3. **Read the report.** Results appear as a stack of sections you can fold and unfold. Tables give the numbers; plots show the picture.
4. **Click on points.** Selecting points in one plot highlights the same rows in every other plot and in the table, so you can see where an odd value came from.



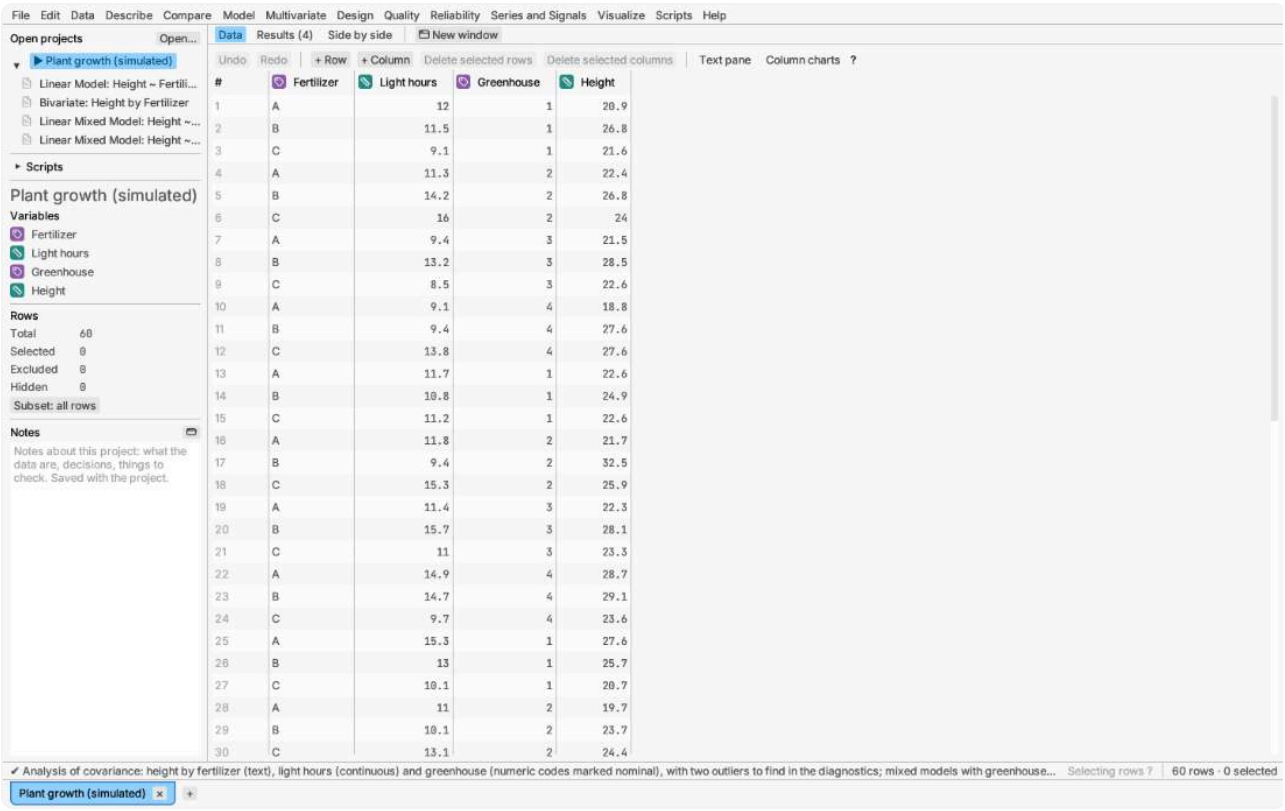
Where to go next

- Your first five minutes walks you through a whole analysis on a sample table.
- Finding your way around names the parts of the window.
- Which analysis do I need? starts from your question and points you to the right menu item.
- The **Menus** chapter has a page per menu, describing every item.
- The Glossary explains the statistical words you will meet in reports.

Your first five minutes

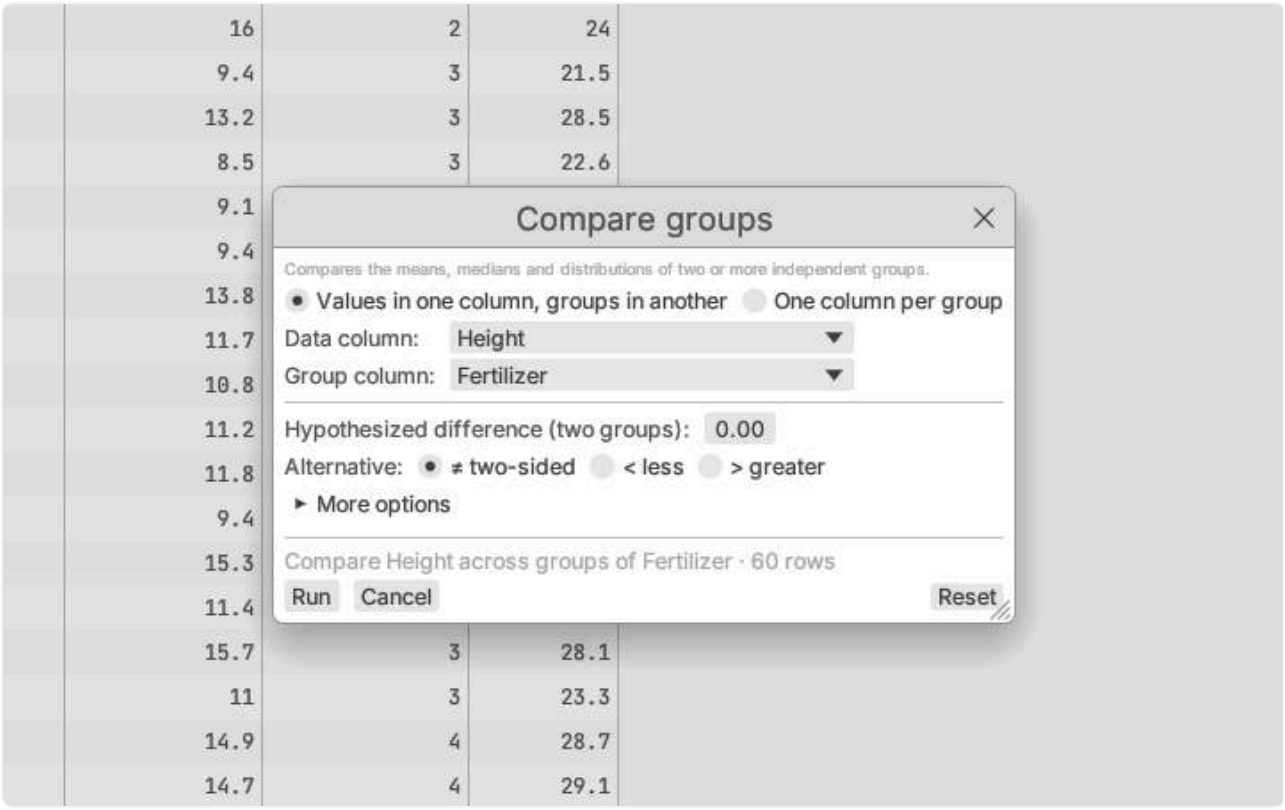
This walk-through uses a sample table of plant heights, so you do not need any data of your own.

1. **Open the sample.** Choose **File ▸ Open sample data ▸ Models ▸ Plant growth**. The table opens, with its columns listed on the left. Like most samples it comes with a few reports already made; you will make your own.

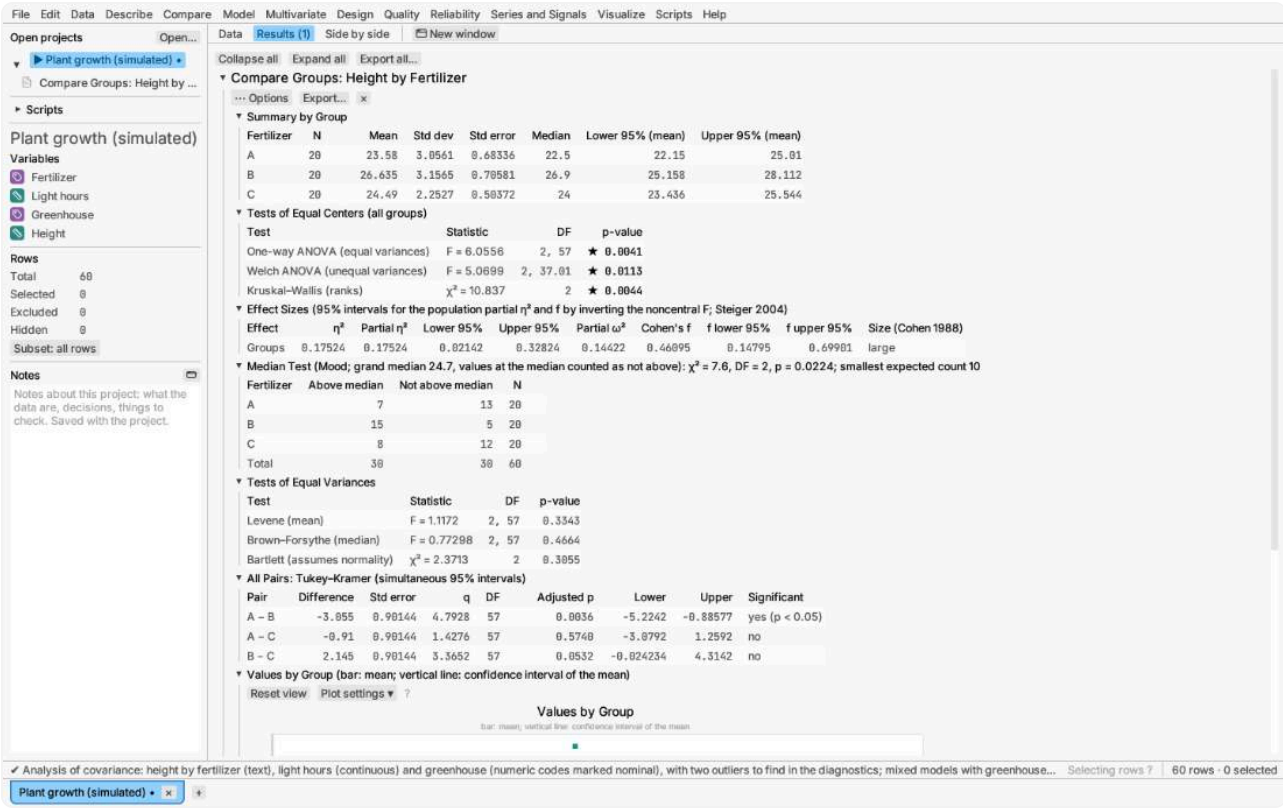


2. **Ask a question.** Do the fertilizers grow plants of different heights? That is a comparison of groups, so choose **Compare ▸ Two or more groups…**.

3. **Fill in the dialog.** Choose `Height` as the data column and `Fertilizer` as the group column. The grey line near the bottom of the dialog says what will run, or what is still missing. Click **Run**.



4. **Read the report.** The report appears at the top of the results, beside the table (the **Results** tab above shows the reports alone). The *Summary by Group* table gives each fertilizer's average height. The *Tests of Equal Centers* table asks whether those averages differ by more than chance: a small *p-value* (below 0.05, marked ★) says they do. Further down, *All Pairs* says which fertilizers differ from which, and a plot shows every plant's height by group. See the Glossary for the words in the report.



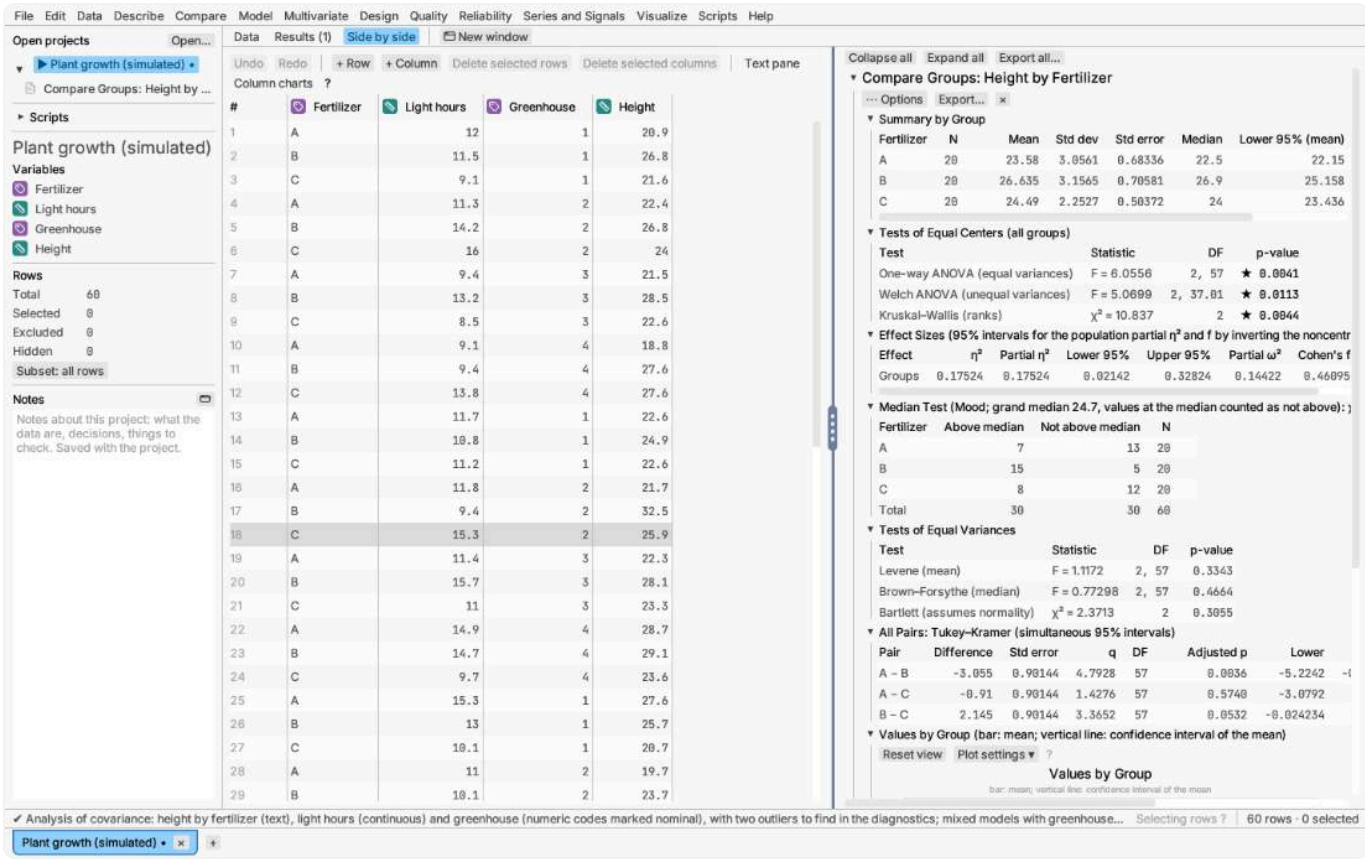
5. **Explore.** Scroll down to the plot and drag a box around the tallest plants. The same rows are highlighted in the table beside the report.

6. **Keep it.** Choose **File ▸ Save project as…**. A project file keeps the data, your reports and what you selected, and reopens exactly as you left it.

That is the pattern for every analysis: open data, choose a menu item, fill in the dialog, read the report.

Finding your way around

When nothing is open, the window shows the **start page**: buttons to make a new table, open a file, import or paste data; your recent projects with their notes; six cards for common tasks ("What do you want to do?"; with no table open, a card first asks you to open a file, start an empty table or pick a sample, then opens its dialog); and links to the sample data and the manuals, with a tip of the day.



- **Menu bar** (top). Every command lives here, grouped by what you want to do. The Menus chapter goes through them one by one.
- **Sidebar** (left). Your open tables, each with its reports and graphs listed under it (click one to go to it), then the columns of the current table. To tidy a project, right-click its name ▸ **New folder** and drag tables, reports or dashboards onto the folder; click a folder to bring its windows to the front. The small icon beside each column says what kind of values it holds: a ruler for measurements, a tag for categories, a numbered list for ordered categories. Under the columns, the row counts say how many rows there are and how many are selected or left out of analyses. Below them, **Notes** is a free-text box saved with the project, for reminders about where the data came from or what you found.

- **Workspace** (middle). Three tabs along the top: **Data** shows the table, **Results** shows the reports, and **Side by side** shows both.
- **Table tabs** (very bottom). One tab per open table. Click to switch, × to close.
- **Status bar** (just above the table tabs). What just happened, and a reminder of how to select rows in plots.

A few habits that help

- **Hover over things.** Most buttons and menu items explain themselves in a tooltip.
- **Right-click things.** Columns, report sections and plots all have menus of their own: rename a column, save a plot as a picture, copy a table.
- **Put a plot in a document or slide.** Right-click the plot and choose **Copy plot**, then paste: it arrives as a sharp picture (on a Mac, as a vector PDF). **Save plot as…** lets you set its size on paper (for example 89 mm, one journal column), a serif font, bigger text or a clear background; the plot remembers those choices.
- **Fold what you are not reading.** Click any report heading to fold it away; **Collapse all** folds everything.
- **Undo.** ⌘Z (Ctrl+Z on Windows and Linux) undoes edits to the table.
- **Resize plots.** Drag the strip under a plot to make it taller, the strip on its right edge to make it wider, or the corner for both. Double-click the strip to go back to the usual size.
- **Change how a plot's axes look. Plot settings ▾** above each plot sets ranges, ticks, grid lines and titles. Its **Ticks** group turns tick marks inward or outward and lets you write your own tick labels (type `\pi` for π); **Spine** moves an axis line away from the data. You can also double-click a number on an axis to rename it, or right-click the axis ▸ **Edit tick label…**. On scatter plots, **Axis breaks ▸ Leave out** cuts a stretch out of an axis, so a few far-off values don't squash the rest; **Add a break** cuts another, and **Segments and marks** puts a piece on a log scale or gives it a fixed share of the axis.
- **Give plots the same look.** Right-click a plot you have formatted and choose **Copy plot style**, then right-click another plot ▸ **Paste plot style**: tick what goes across (ranges, ticks and grid, font and size, colours and markers, reference lines, key) and paste onto that plot, every plot in the report, or every plot in the project.
- **The ⋯ Options button** on each report adds or removes extra tables and plots without running the analysis again. Its **Duplicate on another table…** puts the same report, with its look, on another open table.
- **Drop a column on an axis.** Drag a column from the variables list onto the axis of a report's plot that shows the analysed column, and the report re-runs on the new column.

- **Filter one report.** ⋯ Options ▸ Filter rows… narrows a report to chosen rows; its **Filter panel** can **Step through** a column one level at a time, or play the levels like a slide show.
- **Look for ▾ at the bottom of a list.** It means the list goes on: scroll to see more choices. A scroll bar beside a list, report or table also means there is more to see.

On a Mac, shortcuts use ⌘; on Windows and Linux they use Ctrl. **Help ▸ Keyboard shortcuts** lists them all.

Which analysis do I need?

Start from your question. Each answer names the menu item to use; the Menus pages say more about each one.

"What do my numbers look like?"

- One column of measurements: **Describe ▸ Distribution…** (average, spread, a histogram, and whether it looks bell-shaped).
- Two columns against each other: **Describe ▸ Bivariate…** (a scatter plot and the correlation).
- Counts of categories: **Describe ▸ Multi-way table…**.
- Any picture you like: **Visualize ▸ Graph canvas…**.

"Is there a difference?"

- One group against a target value: **Compare ▸ One sample vs a value…**.
- Two or more groups (for example, three suppliers): **Compare ▸ Two or more groups…**.
- The same things measured twice (before and after): **Compare ▸ Paired or repeated measures…**.
- Percentages (pass rates, yes/no answers): **Compare ▸ Proportions…**.
- How many people or samples you need: **Compare ▸ Power and sample size…**.
- How likely a difference is, as a probability rather than a p-value, or evidence that there is *no* difference: the **Bayesian** submenu under **Compare**.

"What drives the result?"

- A measurement explained by one or more other columns: **Model ▸ Linear model…**.
- A yes/no outcome, or a count: **Model ▸ Generalized linear model…**.
- Many possible predictors and you want the useful ones: **Model ▸ Stepwise selection…** or **Model ▸ Penalized regression…**.

- Prediction is what matters, not explanation: the decision tree, forest and boosting items under **Model**.
- A curve that bends or levels off (growth, decay, a dose–response curve): **Model ▸ Nonlinear regression…**, or **Model ▸ Curve templates…** for a laboratory assay such as an immunoassay standard curve or enzyme kinetics.

"Some of my values are missing"

- Fill the gaps with estimates, or make several filled-in copies so that later results allow for the uncertainty: **Data ▸ Impute missing values…**.

"How do many columns relate?"

- Which columns move together: **Multivariate ▸ Correlations…**.
- Boil many columns down to a few: **Multivariate ▸ Principal components…**.
- Find natural groups of rows: the clustering items under **Multivariate**.

"What are people saying?"

- The words and themes in a column of comments: **Multivariate ▸ Text exploration…** or **Multivariate ▸ Topic model…**.
- Whether the comments are positive or negative: **Multivariate ▸ Sentiment scores…**.

"How should I run my experiment?"

Use the **Design** menu. It makes a new table listing the runs to do. You do not need any data to start. After the experiment, fill in the results and analyse them, usually with **Model ▸ Linear model…** or **Model ▸ Response surface (second-order model)…**.

"Is my process stable and capable?"

- Watch a process over time: **Quality ▸ Control chart…**.
- Does it meet its specification limits: **Quality ▸ Process capability…**.
- Can we trust our measuring device: the gauge study items under **Quality**.

"How long until it fails?"

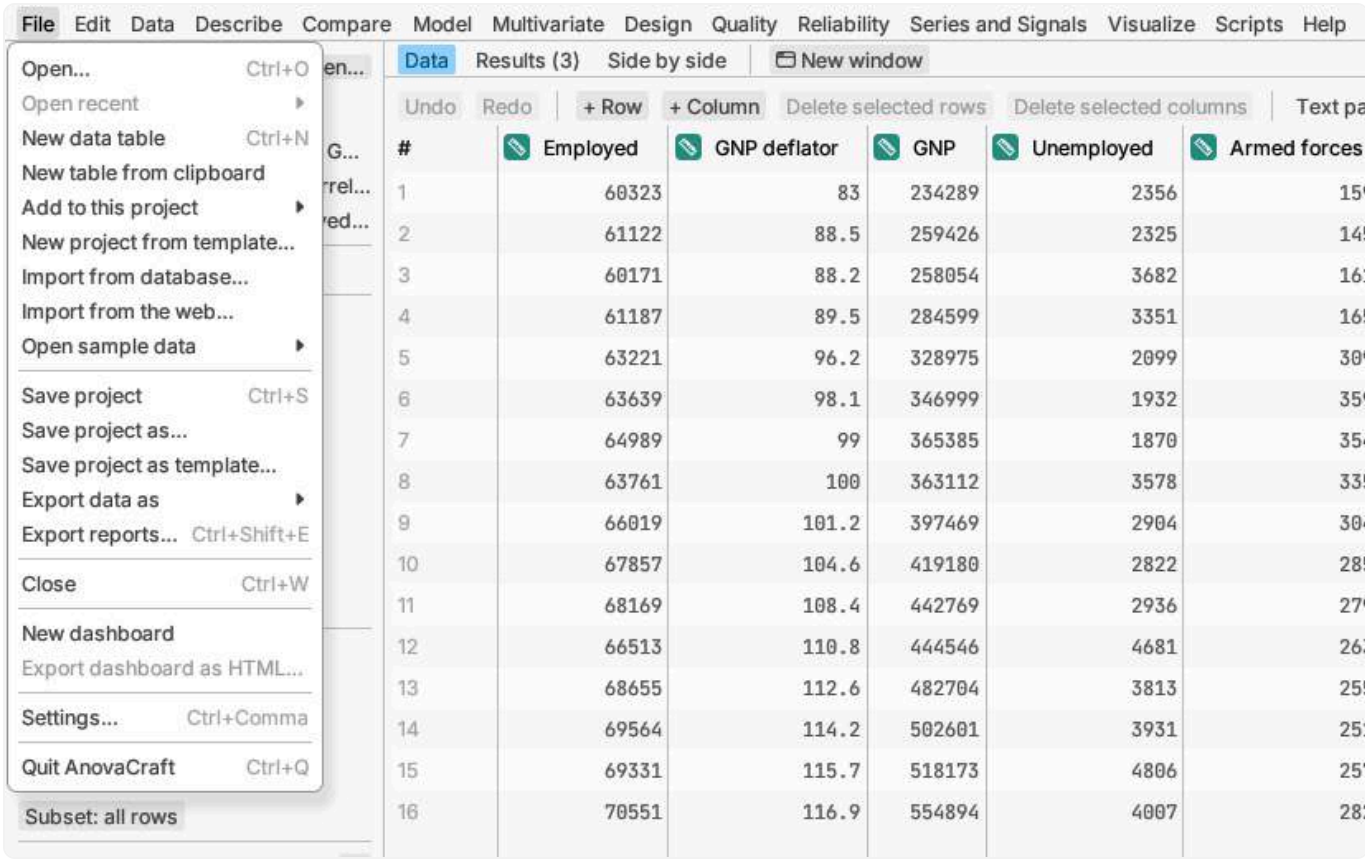
- Time until an event, with some units still running: **Reliability ▸ Survival curves…**.
- Fit a life distribution (such as Weibull): **Reliability ▸ Life distributions…**.

"What happens next month?"

Use the **Series and Signals** menu: describe the series, then forecast it.

File

Opening, saving and sharing your work.



Opening data

Open... opens a data file or a saved project. It reads comma- and tab-separated text, JSON, and spreadsheet workbooks (Excel and OpenDocument; each sheet becomes its own table). You can also drag files onto the window. Text files and workbooks first show a preview of their cells, where you click the cell where the table starts (its first column name) if it does not start in the top-left corner, choose the sheets to open, and press **Open**. Pictures (PNG, JPEG, TIFF, BMP) open as a table with one row per pixel: its position `x` and `y` and its brightness, or red, green and blue. A large picture asks whether to shrink it, cut out a part, or keep every pixel. New data opens showing just the table; no analysis runs until you pick one from a menu. Saved projects and sample data open with their reports, as they were saved.

Open recent lists the files you opened lately.

New data table starts an empty table to type into, like a fresh spreadsheet.

New table from clipboard makes a table from cells you copied in a spreadsheet. The same preview shows first; if the first copied line holds names, they become the column names.

Add to this project puts a new table into the project you are working in instead of starting a project of its own: an **Empty table**, a **Table from clipboard**, or a **Data file…**. The data file preview also has a box, *Add to the open project*, that does the same for any file you open.

New project from template… builds a template's reports, graphs and dashboards on new data: an open table, a data file, or every data file in a folder (each becomes a table of one new project). Columns with the same names match by themselves; for the others you choose which column stands in.

Import from database… reads the result of a database query (SQLite files, PostgreSQL, MySQL or MariaDB servers). Passwords are never saved.

Import from the web… downloads JSON or CSV data, or a table on a web page, from a web address.

Open sample data has dozens of small example tables, grouped by topic. Each one is chosen to show off a kind of analysis and most open with a report already made, so they are the quickest way to learn.

More detail: Documents, files and projects, The Open Data dialog, Data from databases and the web, Sample data.

Saving and sharing

Save project and **Save project as…** save everything in one `.acraft` project file: the tables, your reports and their settings, and which rows are selected or excluded. Reopening the project rebuilds every report exactly.

Save project as template… saves the project's reports, graphs, dashboards, scripts and plot styles without the data, as a `.acrafttemplate` file, so you can build the same reports on next month's data with **New project from template…**.

Export data as saves just the table, without reports, for other programs: **Comma-separated (.csv)** or **Tab-separated (.tsv)** text, **JSON rows (.json)**, or an **Excel workbook (.xlsx)**. Tick **Write value labels** to write names (Female, Male) instead of the codes behind them. **Image (PNG or 16-bit TIFF)…** turns a column back into a grey picture, placing each row by its x and y columns.

Export reports… saves reports as a document to read or send: PDF, Word, PowerPoint, a web page, Markdown or an Excel workbook. **Export all…** at the top of the results does the same.

Close closes the current table (you are asked first if it has unsaved changes). In a project of several tables it reads *Remove table from project*, beside *Close project*.

More detail: Saving plots and results.

Dashboards

New dashboard makes a page of report tiles and notes from the project's tables, with filters that every tile follows. Good for a summary you look at again and again.

Export dashboard as HTML… saves a dashboard as a web page anyone can open in a browser.

Removing a dashboard asks first, and **Edit ▸ Undo** brings it back.

More detail: Dashboards.

The app itself

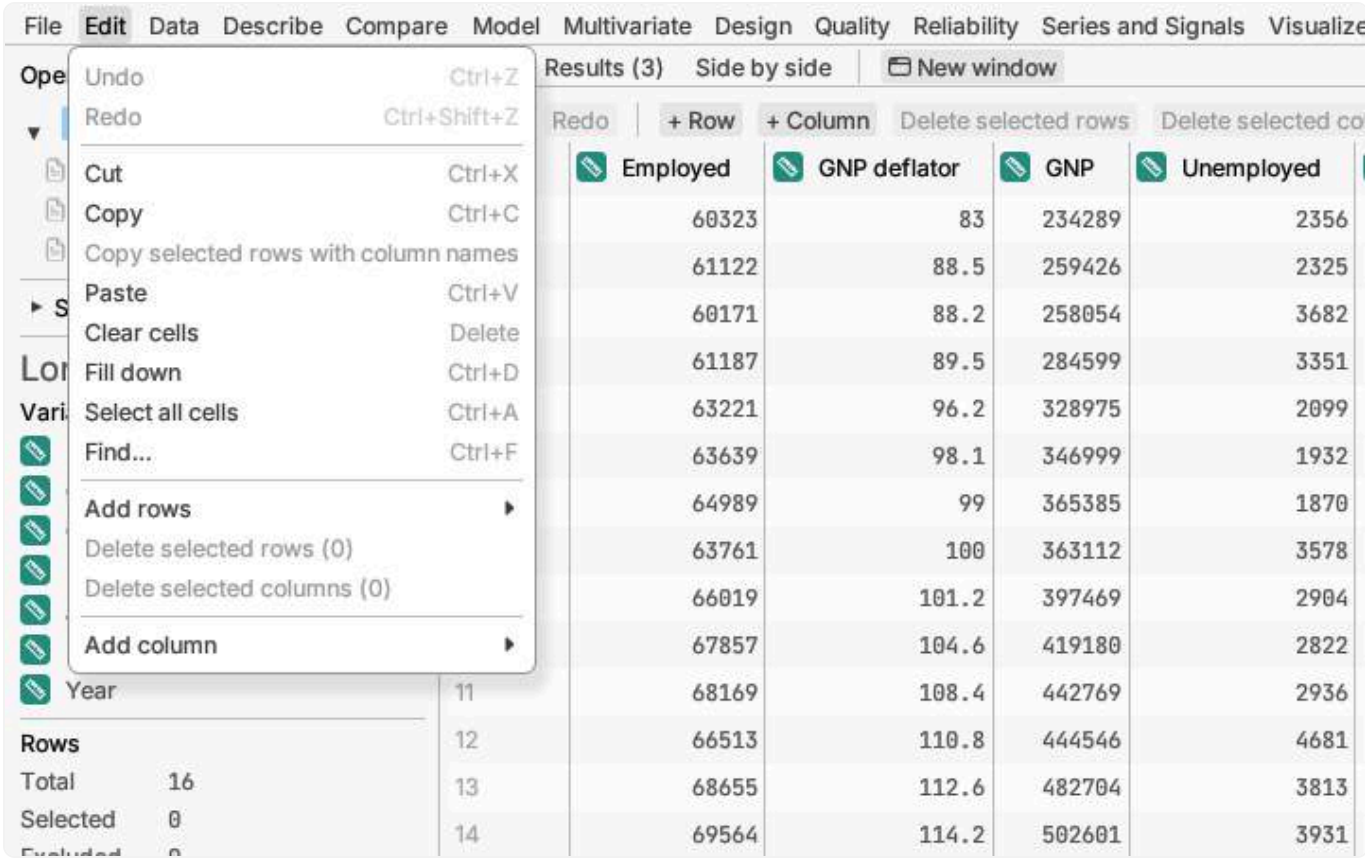
Settings… changes the colour scheme (light, dark, or follow the system), the text size, how long the pointer rests on something before its explanation pops up, how numbers and p-values are shown in reports, whether the getting-started tips appear, and whether simple files open without the preview.

Quit AnovaCraft closes the app, asking first about unsaved work.

More detail: Settings.

Edit

Changing the values in the table, much as you would in a spreadsheet. Click a cell and start typing, or double-click it, to change it. A value too long for its cell opens in a larger box so you can see all of it.



Undo and the clipboard

Undo and **Redo** step back and forward through your edits (the last 500 are kept).

Cut, **Copy** and **Paste** move cells to and from other programs, such as a spreadsheet. **Copy selected rows with column names** copies whole selected rows with a header line, ready to paste elsewhere.

Clear cells empties the selected cells, leaving them missing.

Fill down copies the top cell of a selection into the cells below it.

Select all cells selects the whole table.

Find… searches the table for a value.

More detail: Editing the data grid, Copying and pasting.

Rows and columns

Add rows adds empty rows at the end: set **How many**, then click **Add row at the end** (it says "Add 5 rows at the end" when you ask for five). The grid's **+ Row** button opens the same choice, and can also insert the rows above or below the rows you selected (or the cell you clicked). In the grid, pressing Enter on the last row also adds a row.

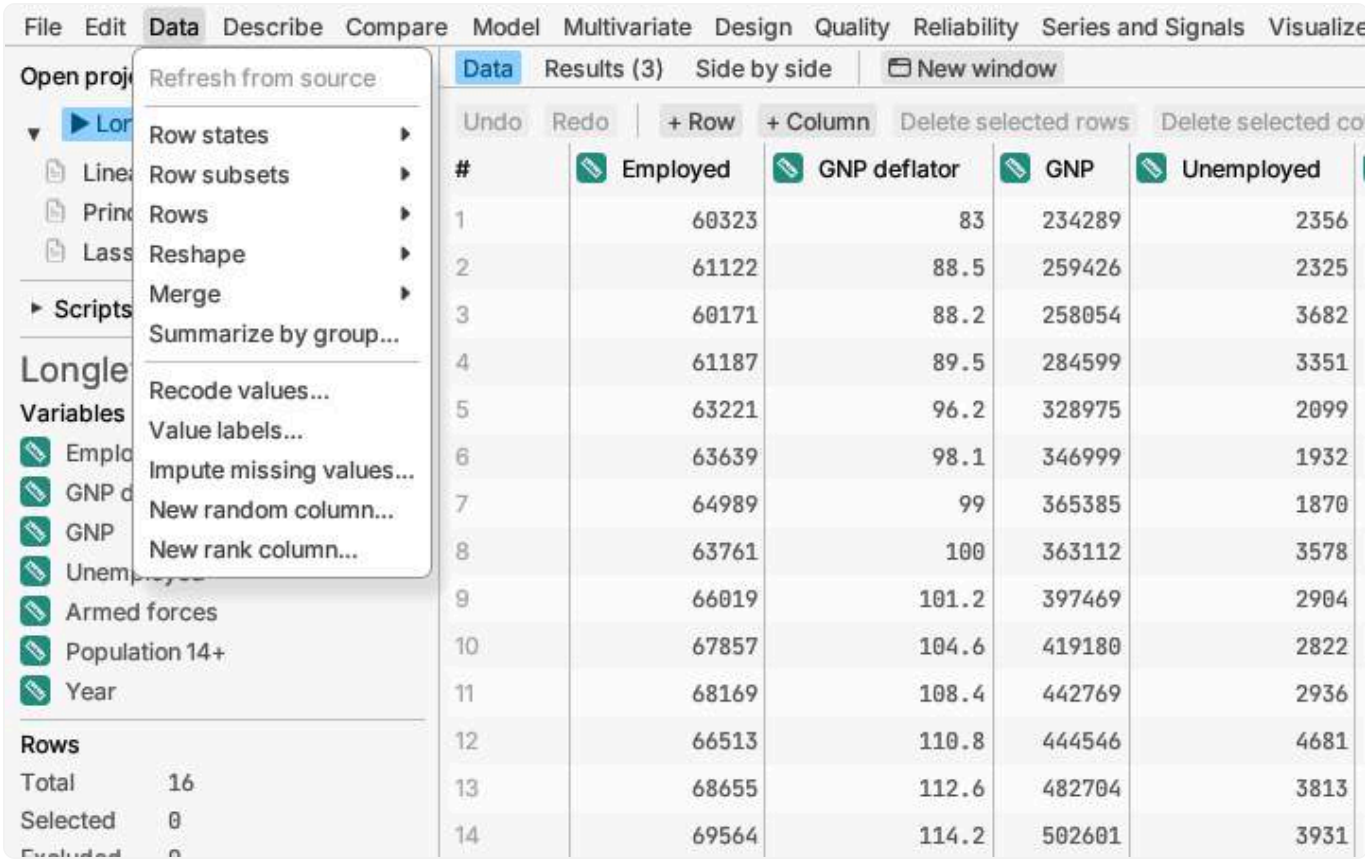
Delete selected rows removes the selected rows (the number in brackets says how many). **Delete selected columns** does the same for the columns whose headings you clicked. You can undo either.

Add column adds a new column holding **Numbers**, **Text** or **Dates**. Change **How many** at the top to add several at once, and type a **Name** if you like (several columns are numbered: Rep 1, Rep 2, …). One undo removes everything you added. The grid's **+ Column** button offers the same, and can also insert the columns to the left or right of the columns you selected (or the cell you clicked), including a formula column. A **Formula…** column is worked out from other columns (for example, weight divided by height squared) and updates itself when the data change. Under the formula, **Reads as** shows it written as maths, with fraction bars, powers and square-root signs, so you can check it. **Random values…** opens the same dialog as **Data ▸ New random column…**.

More detail: Formula columns, Dates and times.

Data

Reshaping, tidying and choosing which rows to use.



Refresh from source reloads a table that came from a database or the web, so it shows the latest data.

Row states

A row can be *selected* (highlighted everywhere), *excluded* (left out of analyses but kept in the table) or *hidden* (left out of plots). Use these to set aside a suspect value without deleting it.

Select all rows, **Clear selection** and **Invert selection** change which rows are selected.

Exclude selected rows from analyses leaves the selected rows out of every report, which updates at once. **Include selected rows in analyses** and **Include all rows in analyses** bring them back. **Select excluded rows** shows you which rows are excluded.

Hide selected rows in plots, **Show selected rows in plots** and **Show all rows in plots** do the same for plots only.

Colour and marker by column… gives every row a colour, and a marker shape, from its value in a column you pick, such as one colour and shape per treatment group. Plots that are not already grouped then draw the rows that way and name the groups in their key. **Clear row colours and markers** takes them off again.

More detail: Row states: select, exclude, hide.

Row subsets

A *row subset* is a named set of rows you can switch to, such as "Men", "Batch 3" or "After the change". While one is active, every report and plot uses only its rows; switch back to **All rows** to use them all.

Save selection as subset… names the selected rows. The menu then lists **All rows** and each saved subset with its row count; pick one to make it active. With a subset active you can also **Select the subset's rows**, **Replace with the selection**, **Rename subset…** or **Delete subset**. The same menu opens from the *Subset* line under the row counts in the sidebar.

More detail: Row subsets.

Rows

Sort… puts the rows in order of one or more columns.

Remove duplicates… keeps the first of each set of repeated rows (you can undo it, or make a new table instead).

Subset and sample… makes a new table of some of the rows: the selected ones, the ones matching a condition, or a random sample.

Use row as column names turns the row you clicked into the column names, for a table whose names ended up as its first row of values. Undo puts the row back.

Reshape

Stack columns… turns several side-by-side columns (say, a column per month) into one long column with a label saying where each value came from. **Unstack by label…** does the reverse. **Transpose…** swaps rows and columns.

Distance matrix… makes a new table of how different every pair of rows is, by a measure you choose (Gower handles a mix of numbers and categories). Use it for **Principal coordinates…**, **Dimension reduction…** or **Clustering…**.

Merge

Combine rows… puts tables with the same columns one under the other. **Join…** matches rows of two tables on a shared key column (such as a customer number) and puts their columns side by side.

Other tools

Summarize by group… makes a new table of statistics (counts, means and so on) for each group, such as the average height per fertilizer.

Recode values… changes values or merges categories, for example turning "NY", "N.Y." and "New York" into one spelling.

Value labels… gives names to codes, such as 1 = Female and 2 = Male. The table keeps the numbers, but the grid, reports and plots show the names. The **Stored values** button above the grid shows the numbers again.

Impute missing values… fills gaps in the data with sensible estimates, so methods that need complete rows can use more of them. Its *multiple imputation* methods make several filled-in copies instead, stacked in a new table; analyses of that table combine the copies, so their results allow for not knowing the missing values.

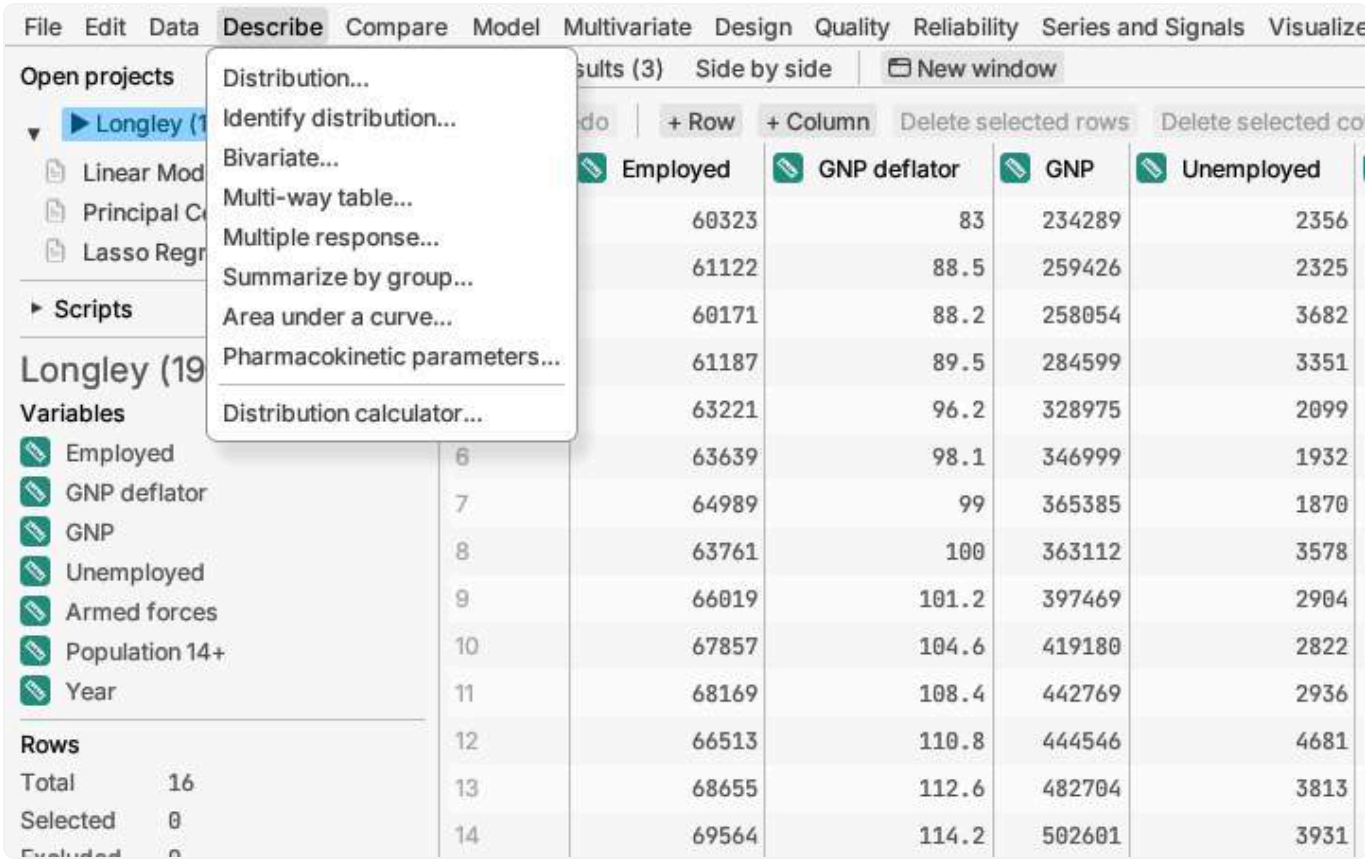
New random column… adds a column of random numbers from a chosen distribution, for simulations or random assignment.

New rank column… adds a column numbering the values of a column from smallest (1) to largest, optionally within each group, with a choice of how equal values are numbered (shared average, lowest, highest, no gaps, or in row order).

More detail: Table operations, Rank columns, Missing values and imputation, Multiple imputation.

Describe

Getting to know your data before testing anything: what is typical, how spread out it is, what shape it has.



Distribution... describes one column at a time. For measurements you get the average, spread and percentiles, a histogram, a box plot and a check of whether the values follow a bell curve. For categories you get counts and percentages. This is usually the first thing to run on new data. *Try it:* sample *Michelson light speed*. More detail

If some values are only known to be below a limit (lab results reported as "< 2", called non-detects) or above one, choose how they are marked under *Censored values* in the same dialog. The report then estimates the distribution and its percentiles without pretending those values are exact. *Try it:* sample *Well water arsenic*. More detail

Identify distribution... tries fourteen common distribution shapes (normal, lognormal, Weibull and others) on one column and ranks how well each fits. Useful when you need to know the shape, for example before working out how often a part will fall out of specification. *Try it:* sample *Seal strength*. More detail

Bivariate… looks at two columns together: a scatter plot and the correlation for two measurements (for a fitted line, use **Model ▸ Linear model…**), a side-by-side comparison for a measurement and a category, or a table of counts for two categories. *Try it:* sample *Norris calibration*. More detail

Multi-way table… counts rows by several categories at once (for example, outcome by treatment within each clinic), with percentages, subtotals and averages. *Try it:* sample *Multi-centre trial*. More detail

Multiple response… counts the answers to a "tick all that apply" question, such as "How did you hear about us?". The answers can be in one column as lists (Email; Phone) or in one yes/no column per choice. You see how many people ticked each choice, and, with a group column such as age group, whether the groups tick choices differently. *Try it:* sample *Visitor survey*. More detail

Summarize by group… makes a table of counts, means and other statistics for each group; it is the same command as **Data ▸ Summarize by group…**.

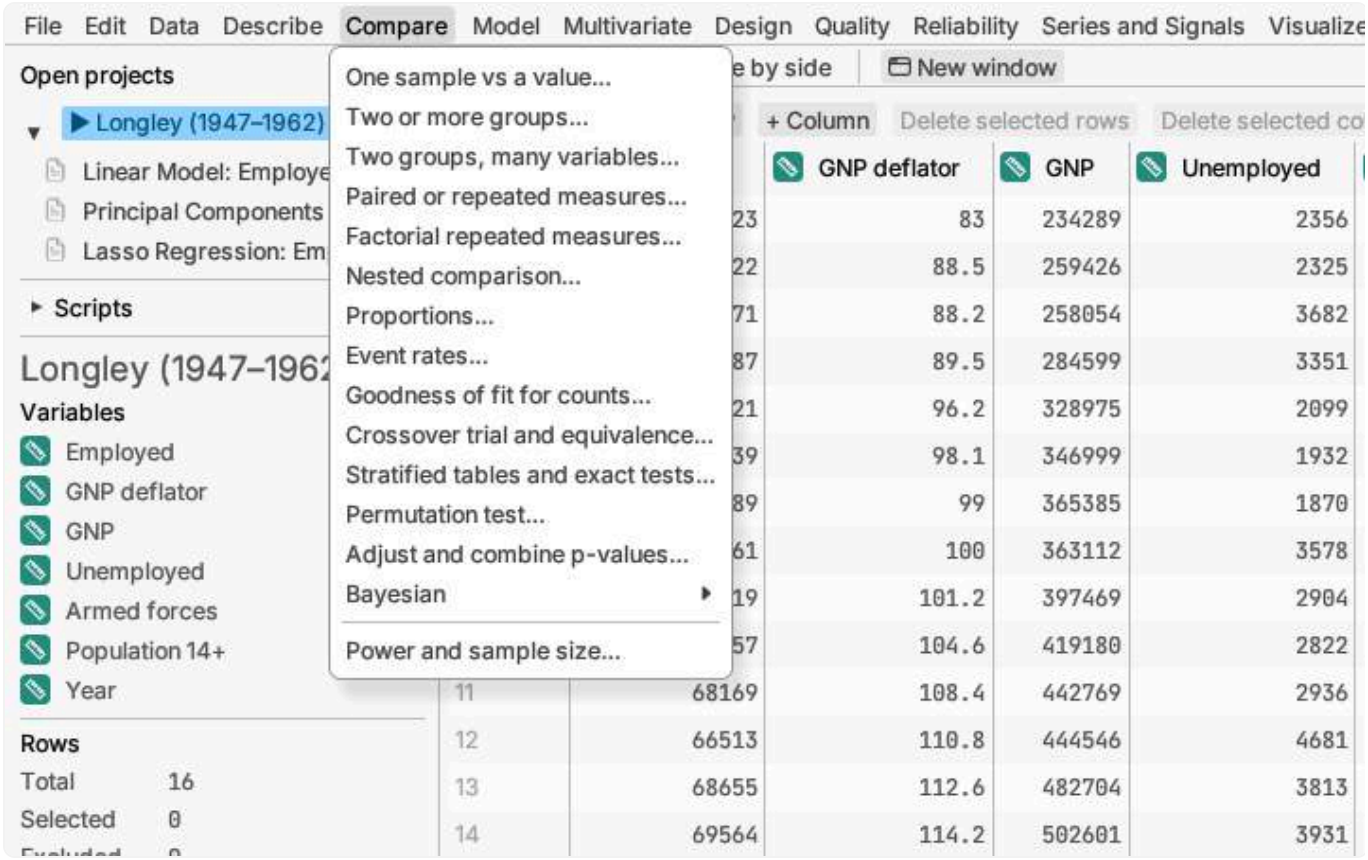
Area under a curve… measures the area under measurements taken over time (or dose, or wavelength), such as a drug's concentration in the blood after a dose. It joins the average at each time with straight lines, gives the area above a baseline with its margin of error, and finds the peaks. It can also save the measurements rescaled to 0–100% or as a share of the total. *Try it:* sample *Plasma concentration*. More detail

Pharmacokinetic parameters… summarises how a drug's concentration in the blood rises and falls after a dose, for each subject: the peak and when it happens, the area under the curve, the half-life, and (given the dose) clearance and volume, then averages them by treatment. *Try it:* sample *PK profiles*. More detail

Distribution calculator… works out probabilities from a named distribution, using no data from the table, such as "what share of a normal distribution lies above 2?". It draws the curve and shades the area. More detail

Compare

Testing whether there is a real difference: between a group and a target, between groups, or before and after. Each report gives the size of the difference with a confidence interval and a p-value (see the Glossary).



One sample vs a value... asks whether a column's average (or median, or spread) differs from a target, such as "do our bags weigh 500 g on average?". It can also show that a value is *close enough* to a target (equivalence). *Try it:* sample *Michelson light speed*. More detail

Two or more groups... compares a measurement across groups, such as three suppliers or two teaching methods. It runs the usual tests (t test, analysis of variance) and versions that do not assume a bell curve, checks whether the groups are equally spread, and says which pairs of groups differ. *Try it:* sample *Silicon resistivity* or *Plant growth*. More detail

Two groups, many variables... runs the same two-group test on many columns at once, such as thirty genes measured in treated and untreated samples. With many tests some p-values are small by chance, so it also gives adjusted p-values and a volcano plot (size of

difference against significance) that picks out the columns worth a closer look. *Try it:* sample *Gene screen*. More detail

Paired or repeated measures… is for the same people or items measured more than once, such as before and after a treatment. Pairing removes the differences between individuals, so it finds smaller effects. *Try it:* sample *Sleep drug trial*. More detail

Factorial repeated measures… is for people or items measured under every combination of two or three conditions (task and cue, time and dose), possibly in separate groups such as age groups. It tests each condition, each group and their combinations, checks the assumption of equal spread with corrections when it fails, and can test one factor within each level of another. *Try it:* sample *Reaction times by task and cue*. More detail

Nested comparison… compares groups when each subject gives several measurements, such as several cells from each mouse. It counts subjects, not measurements, so the test is not fooled by many measurements of a few subjects, and shows how much the subjects differ. *Try it:* sample *Dendrite lengths per mouse*. More detail

Proportions… compares percentages, such as pass rates or survey "yes" answers, against a target or between groups. *Try it:* sample *Customer churn*. More detail

Event rates… compares how often events happen per unit of time or exposure, such as accidents per thousand hours. More detail

Goodness of fit for counts… checks whether counts in categories match expected proportions (are the dice fair?), or whether counts follow a Poisson pattern. *Try it:* sample *Horse-kick deaths*. More detail

Crossover trial and equivalence… analyses a study in which each subject gets both treatments in turn, and checks whether two products are equivalent (for example, a generic medicine). *Try it:* sample *Bioequivalence crossover*. More detail

Stratified tables and exact tests… compares yes/no outcomes between two groups across several strata (such as clinics), and gives exact tests for small tables of counts. *Try it:* sample *Multi-centre trial*. More detail

Permutation test… compares two groups, or tests a correlation, by shuffling the data many times instead of assuming a bell curve. A good choice for small or oddly shaped samples. *Try it:* sample *Repair times*. More detail

Adjust and combine p-values… takes a column of p-values from tests you have already run (one per row) and adjusts them for the number of tests, by the usual methods

(Bonferroni, Holm, false discovery rate and others). It also combines them into one overall test: is there any effect at all? *Try it:* sample *Trial endpoints*. More detail

The **Bayesian** submenu (also under Model) holds four analyses that describe what the data say as probabilities, starting from a stated prior belief, instead of p-values.

Bayesian estimation… gives the likely range of a mean, a proportion or an event rate (or of the difference between two groups) as a probability distribution, and answers questions such as "how likely is the new treatment to be better by at least 5 points?". When a group had no events, the average of a ratio does not exist and shows as —; read the median and the intervals instead. *Try it:* sample *Response rates by site*. More detail

Bayes factor tests… weigh the evidence for a difference or a correlation against the evidence for none. Unlike a p-value, a Bayes factor can also show that there is probably *no* effect. *Try it:* sample *Stroop reaction times*. More detail

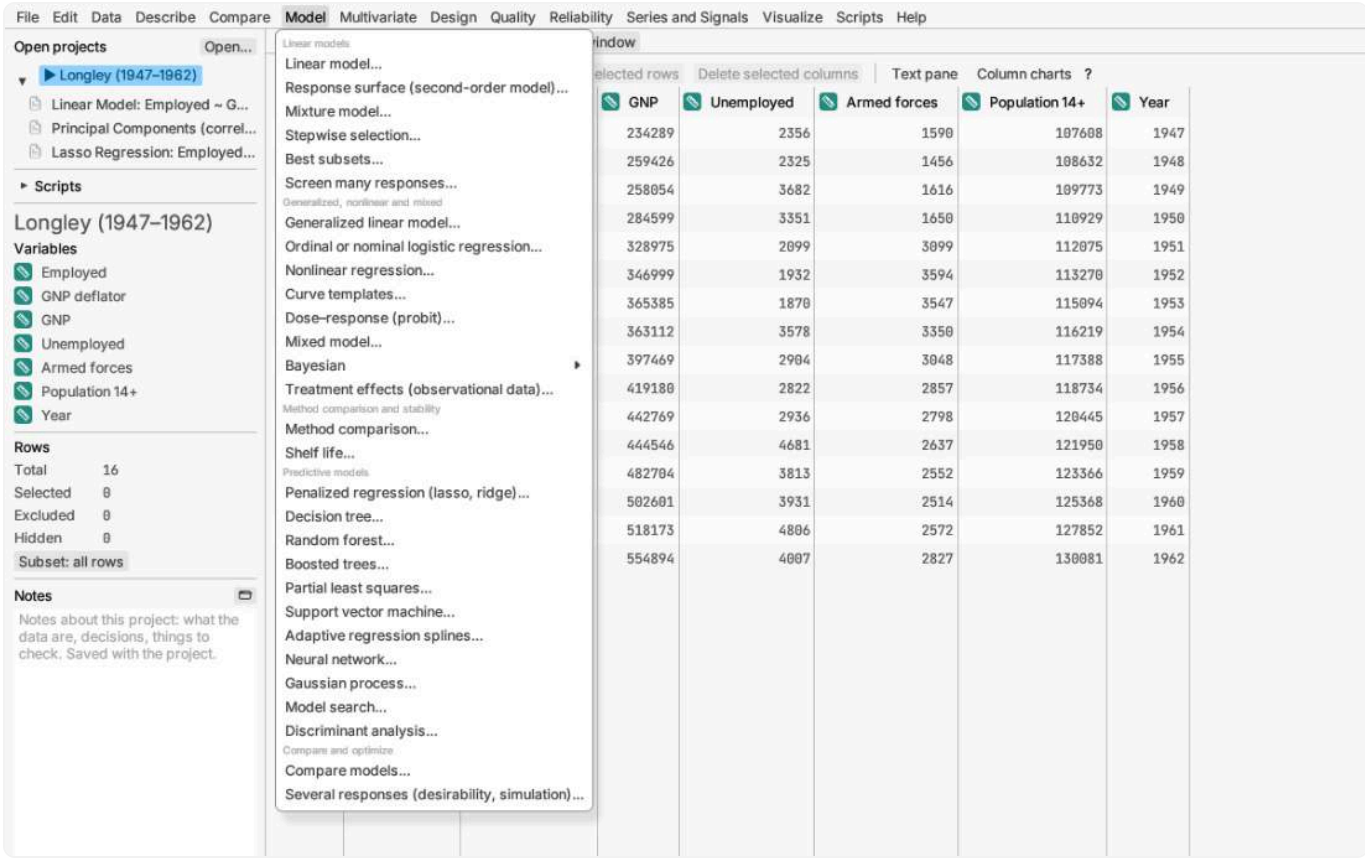
Bayesian linear model or ANOVA… compares every combination of the chosen effects and says how likely each effect is to matter. More detail

Bayesian regression or hierarchical model… fits regression models (including yes/no outcomes and counts) for grouped data, such as patients within hospitals, by simulation, and checks that the simulation worked. *Try it:* sample *Eight schools*. More detail

Power and sample size… helps you plan before collecting data: how many measurements you need to detect a difference of a given size, or how likely a planned study is to find it. It also tells you how many values a tolerance interval needs, including one that assumes nothing about the shape of the data. It uses no data (any open table will do, even a new empty one). More detail

Model

Explaining or predicting one column (the *response*) from others (the *predictors* or *factors*). A model is an equation; the report says how well it fits, which predictors matter, and lets you try "what if" settings.



Linear models

Linear model... is the workhorse. It explains a measurement from one or more other columns, which can be measurements or categories, and from their combinations. The report gives the equation, how much of the variation it explains (R^2), which predictors matter, a plot of the fit, and a Response Explorer where you drag each factor's setting and watch the prediction change. Plots of what is left over (residuals) are in the report's ⋯ Options menu. Analysis of variance and covariance are linear models too. Under *More options* a measured predictor can bend (a spline term), and the report's ⋯ Options can compare one factor's levels at each level of another (comparisons within slices) or each group's slope. *Try it:* sample *Longley economy*. More detail

Response surface (second-order model)… fits a curved model with squared terms and finds the settings that give the highest or lowest response, with coloured contour maps of the surface. Used after a designed experiment. *Try it:* sample *Reaction yield*. More detail

Mixture model… is for blends whose ingredients add up to 100%, such as a recipe or an alloy, and draws the triangle plot of the blend. *Try it:* sample *Yarn elongation*. More detail

Stepwise selection… chooses which of many candidate predictors to keep, adding or dropping one at a time. Tick *Several responses* to run a separate search for each of several responses and explore them together. *Try it:* sample *Hald cement*. More detail

Best subsets… fits every combination of the candidate predictors and lists the best model of each size, so you can weigh simplicity against fit. *Try it:* sample *Hald cement*. More detail

Screen many responses… fits the same model to many response columns at once, such as the counts of thousands of genes against a treatment, and lists which responses the factors affect, with p-values adjusted so that few of those listed are false alarms. A volcano plot shows each response's effect against its strength of evidence; any response's full report opens from ⋯ Options. *Try it:* sample *Transcript counts*. More detail

Generalized, nonlinear and mixed

Generalized linear model… handles responses a linear model cannot: yes/no outcomes (logistic regression), counts (Poisson regression), and positive skewed measurements. *Try it:* sample *Customer churn*. More detail

Ordinal or nominal logistic regression… predicts a category with more than two levels: ordered ones, such as a rating from poor to excellent, or unordered ones, such as the way people travel to work. *Try it:* samples *Service ratings* and *Commute modes*. More detail

Nonlinear regression… fits a curve whose shape you choose, such as growth that levels off, a dose–response curve or a peak. It can also fit several groups at once and test whether they differ, read unknown samples off a standard curve, set aside outliers, and fit a rate equation (differential equation). *Try it:* samples *Seedling growth*, *Compound potency*, *Assay standard curve* and *Substrate depletion*. More detail

Curve templates… fits a ready-made curve for a laboratory assay: an immunoassay standard curve, receptor binding, enzyme kinetics and inhibition, a dose–response curve, or growth, decay and survival. Choose the kind of experiment and the template, then your columns; the starting values come from your data. Besides the fit, the report gives what the

assay needs, such as Ki, kcat, EC50 or a half-life, a check of the standards against acceptance limits with unknown samples read off the curve, or the best-supported inhibition mechanism. *Try it:* samples *ELISA plate*, *Enzyme inhibition*, *Antagonist shift*, *Potency assay* and the other curve template samples. More detail

Dose–response (probit)… finds the dose that affects a given share of subjects, such as the dose that kills half the insects. *Try it:* sample *Rotenone*. More detail

Treatment effects (observational data)… estimates what a treatment did when it was not given at random, by comparing treated and untreated rows that were alike before treatment (matching or weighting on each row's chance of being treated). It shows whether the groups were made alike (balance), the effect with its interval, and the effect within subgroups. *Try it:* sample *Cardiac rehabilitation*. More detail

Mixed model… is a linear model for grouped data, such as several measurements per patient or per batch, where the groups are a sample from a larger population. It also handles ordered ratings (choose the *Ordinal* family), rows that stand for several observations (*Frequency*) or count more than others (*Weight*), groups whose scatter differs (*Residual variance differs by*) and robust standard errors. *Try it:* sample *Furnace coatings*, or *Wine bitterness* for ratings. More detail

The **Bayesian** submenu is also under Compare: **Bayesian estimation…**, **Bayes factor tests…**, **Bayesian linear model or ANOVA…** (every combination of effects compared, with how likely each effect is to matter) and **Bayesian regression or hierarchical model…** (regression for grouped data, such as patients within hospitals, fitted by simulation; *try it:* sample *Eight schools*). See Compare and the Reference Manual.

Method comparison and stability

Method comparison… checks whether two measuring methods agree, allowing for error in both. If one method rises while the other falls, the report notes that the robust (Passing–Bablok) line cannot give a full range for its slope. *Try it:* sample *Glucose methods*. More detail

Shelf life… estimates how long a product stays within specification from stability-test data. With only two test times per batch it cannot check whether batches degrade at different rates, so it assumes they do not and says so. If your batches are a sample of future production, choose *Batches are: A random sample of batches* for a shelf life a new batch can be expected to meet. *Try it:* sample *Tablet stability*, or *Capsule stability* for random lots. More detail

Predictive models

These aim to predict well rather than to explain. Most hold back part of the data to check predictions honestly (validation).

Penalized regression (lasso, ridge)... is a linear model that shrinks weak predictors toward zero, which helps when there are many predictors. The lasso can drop predictors entirely. *Try it:* sample *Longley economy*. More detail

Decision tree... splits the data into groups with simple rules ("income above 40,000 and fewer than two loans"), drawn as a tree that is easy to explain. *Try it:* sample *Loan applications*. More detail

Random forest... averages hundreds of trees for better predictions and ranks the predictors by importance. *Try it:* sample *Loan applications*. More detail

Boosted trees... builds trees one after another, each fixing the mistakes of the ones before. Often the most accurate of the tree methods. *Try it:* sample *Loan applications*. More detail

Partial least squares... predicts from very many related columns, such as the readings of a spectrum, by first combining them into a few summaries. *Try it:* sample *Fuel spectra*. More detail

Support vector machine... separates classes (or predicts a measurement) with a flexible boundary. *Try it:* sample *Pump sensors*. More detail

Adaptive regression splines... builds a model from straight-line pieces with bends, finding curves and interactions on its own. *Try it:* sample *Fuel consumption*. More detail

Neural network... learns a flexible prediction from layers of simple units, which can capture curves and interactions no one specified. It checks itself on held-back rows to avoid memorizing the data. *Try it:* sample *Polymer batches (simulated)*. More detail

Gaussian process... fits a smooth surface through a measured response and says how sure it is everywhere: sure near your runs, less sure between and beyond them. It suits small experiments and computer simulations. *Try it:* sample *Reaction yield (simulated)*. More detail

Discriminant analysis... assigns rows to known groups from their measurements, such as telling honey's origin from its chemistry. (Also on the Multivariate menu.) *Try it:* sample *Honey origins*. More detail

Compare and optimize

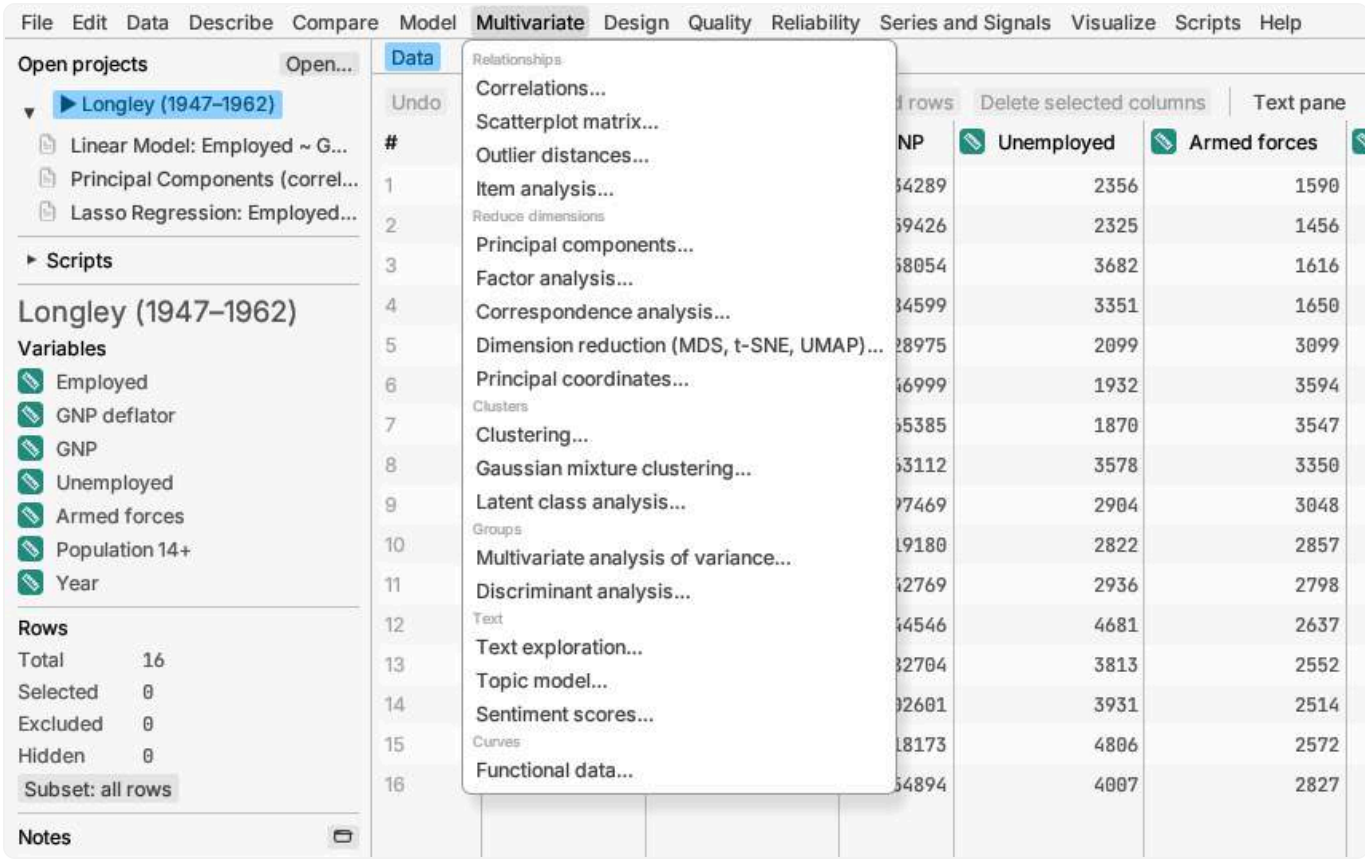
Compare models… puts the predictions of several saved models side by side, with measures of accuracy, to pick the best. *Try it:* sample *Loan applications*. More detail

Model search… fits every suitable predictive model to one response, ranks them by how well they predict the held-back rows, and opens the winner's full report. A quick way to find a good starting model. *Try it:* sample *Polymer batches (simulated)*. More detail

Several responses (desirability, simulation)… handles several responses at once (with the same terms for all, or each response's own terms chosen stepwise), finds settings that balance all your goals, and simulates how variation in the settings spreads to the results. *Try it:* sample *Coating formulation*. More detail

Multivariate

Looking at many columns at once: how they move together, how to sum them up in a few numbers, and how rows fall into groups.



Relationships

Correlations… measures how strongly each pair of columns moves together, from −1 to +1, as a table and a colour-coded map. *Try it:* sample *Flower measurements*. More detail

Scatterplot matrix… draws a scatter plot for every pair of columns in one grid, so patterns and odd rows stand out. More detail

Outlier distances… finds rows that are unusual in their combination of values, even when no single value looks odd. More detail

Item analysis… checks whether the questions of a questionnaire or test measure the same thing consistently (Cronbach's alpha). *Try it:* sample *Course evaluation*. More detail

Reduce dimensions

Principal components… combines many related columns into a few new ones that keep most of the information, and plots rows and columns on them. *Try it:* sample *Flower measurements*. More detail

Factor analysis… looks for a few hidden traits behind many observed scores, such as verbal and numerical ability behind a set of tests. *Try it:* sample *Ability tests*. More detail

Correspondence analysis… draws a map of a table of counts, placing categories that go together close to each other. *Try it:* sample *Customer survey*. More detail

Dimension reduction (MDS, t-SNE, UMAP)… draws a two-dimensional map in which similar rows sit close together, from many columns or from a table of distances. *Try it:* samples *Expression profiles* and *Town distances*. More detail

Principal coordinates… draws a map of the rows from any measure of how different they are, including one for a mix of numbers and categories (Gower), or from a table of distances, and shows how much of the differences each axis captures. *Try it:* samples *Shrub traits* and *Town distances*. More detail

Clusters

Clustering… sorts rows into groups of similar rows, either a chosen number of groups (k-means) or a family tree of groups (hierarchical). Not sure how many groups? k-means can compare several numbers of groups and suggest one. If you already have a guess at the group centers, type them in (or read them from a table) and k-means starts from there. The family tree can also use other distances, such as Gower for columns that mix numbers and categories, or a table of distances. More detail

Gaussian mixture clustering… finds groups that may overlap and have different shapes, and gives each row a probability of belonging to each. *Try it:* sample *Cell populations*. More detail

Latent class analysis… finds hidden groups from yes/no or category answers, such as types of patients from a symptom checklist. *Try it:* sample *Symptom checklist*. More detail

Groups

Multivariate analysis of variance… compares groups on several measurements at once. *Try it:* sample *Plastic film*. More detail

Discriminant analysis… learns how known groups differ and assigns new rows to a group. *Try it:* sample *Honey origins*. More detail

Curves

Functional data… treats each row group's readings over time (or another scale) as one smooth curve, such as a temperature profile per batch. It smooths the curves, can line up their peaks, finds the main ways they differ from each other, and can relate those to factors such as settings in an experiment. *Try it:* sample *Exotherm profiles*. More detail

Text

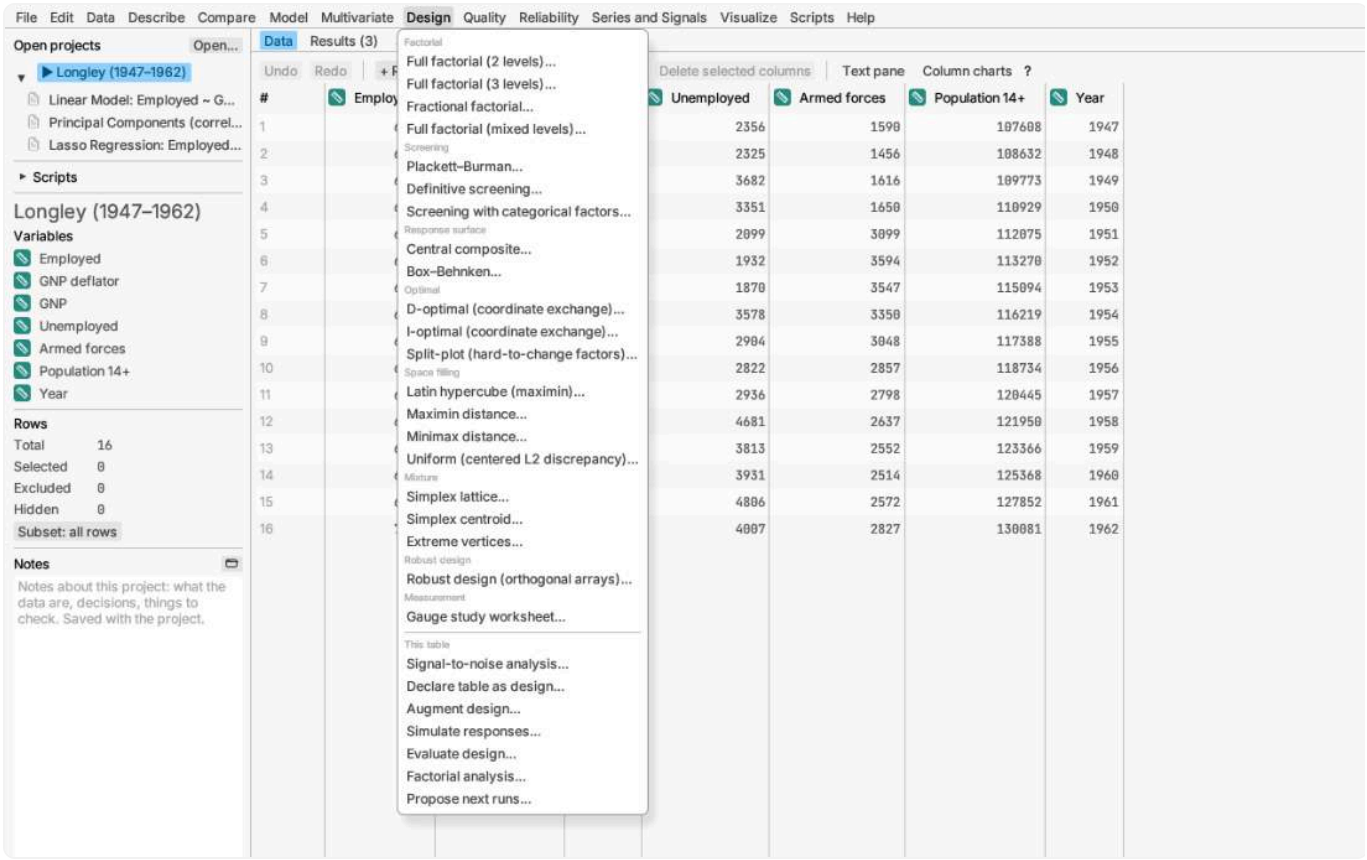
Text exploration… breaks a column of free text (comments, reviews, notes) into words and short phrases, counts how often each appears, and shows the most common terms in a table and a word cloud. It can also find the main themes behind the wording and save them as columns for other analyses. *Try it:* sample *Espresso machine reviews*. More detail

Topic model… finds a chosen number of topics, each a set of words that tend to appear together, and says how much of each document belongs to each topic. *Try it:* sample *Espresso machine reviews*. More detail

Sentiment scores… scores each document from negative to positive using a word list that also handles "not" and "very", and compares the scores across groups. *Try it:* sample *Espresso machine reviews*. More detail

Design

Planning an experiment: which combinations of settings to run so that a small number of runs tells you as much as possible. Each design item makes a new table listing the runs in a random order, with an empty column for each result. You need no data to start. Run the experiment, type in the results, then analyse them with **Model ▸ Linear model…**, **Response surface**, **Mixture model** or **Several responses** (a split-plot design fills in **Model ▸ Mixed model…** for you). The items under *This table* work on a table you already have.



More detail on everything here: Design in the Reference Manual.

Factorial and screening designs

Full factorial (2 levels)… tries every combination of a low and a high setting of each factor. Clear and simple for up to about five factors.

Full factorial (3 levels)… does the same with low, middle and high settings, which shows curvature.

Fractional factorial… runs a well-chosen fraction of the full set (say 16 runs instead of 64), giving up some information on interactions.

Full factorial (mixed levels)… allows a different number of settings per factor (for example, 3 machines × 2 temperatures × 4 materials).

Plackett–Burman… and **Definitive screening…** check many factors in very few runs, to find the handful that matter. Definitive screening also spots curvature; you can add a few extra runs or center runs, and compare the choices, for a more reliable answer.

Screening with categorical factors… does the same when some factors are categories, such as three suppliers or two machines, mixed with numeric ones.

Response surface designs

Central composite… and **Box–Behnken…** are designs for fitting a curved model and finding the best settings, once you know which factors matter. Analyse them with **Model ▸ Response surface (second-order model)…**.

Custom designs

D-optimal (coordinate exchange)… and **I-optimal (coordinate exchange)…** build a design for exactly the model, number of runs and restrictions you have, when no standard design fits. D-optimal is best for estimating effects, I-optimal for predicting.

Split-plot (hard-to-change factors)… is for experiments where one factor, such as an oven temperature, is slow to change, so runs are grouped by it.

Space-filling designs

For computer simulations and other experiments with no random error, these spread the runs evenly through the region: **Latin hypercube (maximin)…**, **Maximin distance…**,

Minimax distance… and **Uniform (centered L2 discrepancy)…**. *Try it:* sample *Borehole water flow*.

Mixture

For recipes whose ingredients add up to 100%: **Simplex lattice…** and **Simplex centroid…** cover the whole range of blends, and **Extreme vertices…** handles lower and upper limits on each ingredient. Analyse them with **Model ▸ Mixture model…**.

Robust design

Robust design (orthogonal arrays)… plans an experiment to find settings that work well despite noise you cannot control, such as humidity. **Signal-to-noise analysis…** analyses it. *Try it:* sample *Moulding shrinkage*.

Measurement

Gauge study worksheet… makes the run sheet for a measurement-system study: which operator measures which part, and when. (Also on the Quality menu.)

This table

Declare table as design… tells the app that a table you already have is an experiment, so it can check it and analyse it like one.

Augment design… adds runs to an existing design, for example to untangle effects that were mixed up or to add a curved term. For a split-plot design it can add new whole plots.

Simulate responses… makes up results for a design from random effects, a few of them large, so you can practise the analysis and see whether the design finds them before you run the experiment.

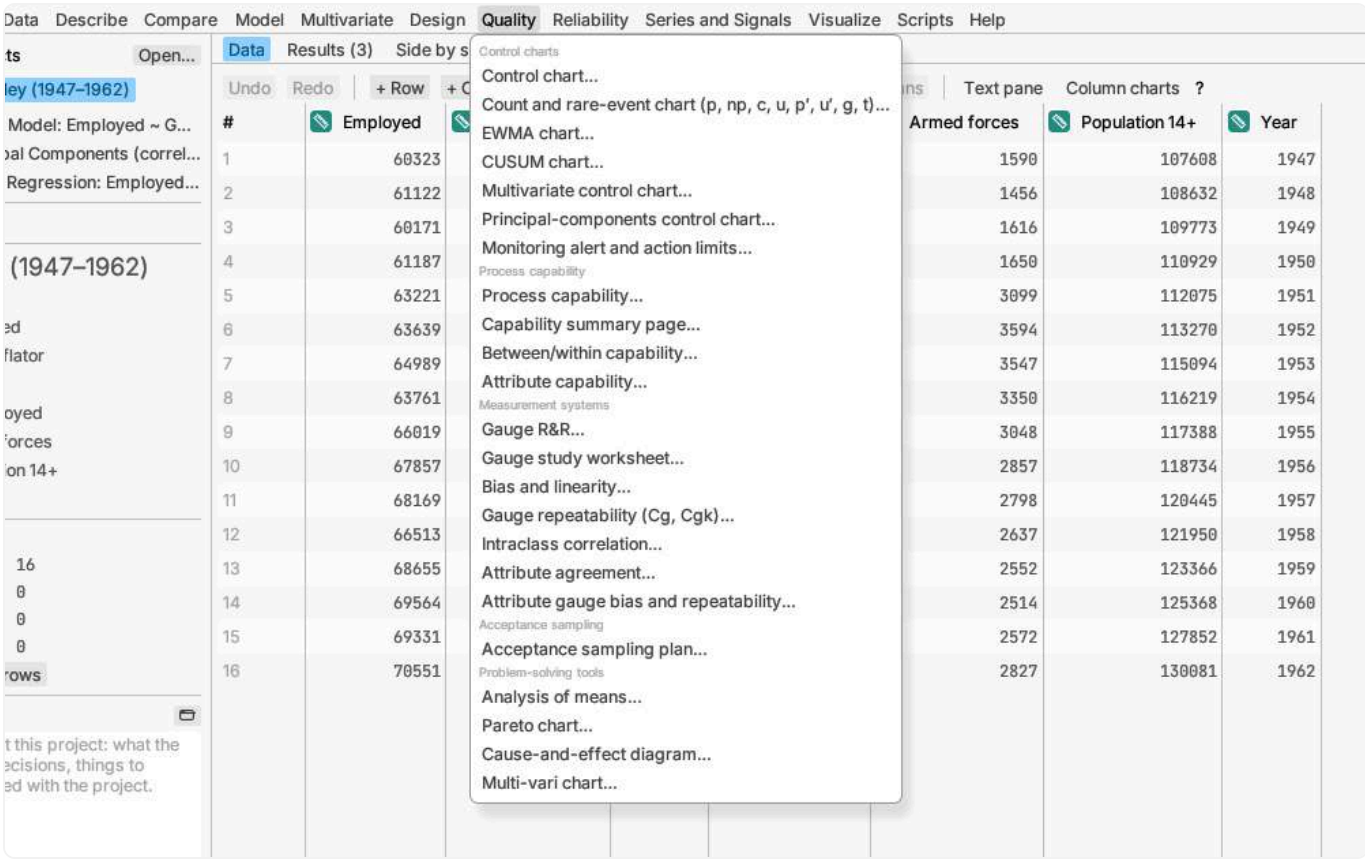
Evaluate design… shows how good a design is before you run it: how precisely it estimates each effect and which effects are mixed up. *Try it:* sample *Design comparison*.

Factorial analysis… analyses a two-level experiment, including one with no repeated runs, and shows which effects stand out from the noise. *Try it:* sample *Filtration rate*.

Propose next runs… looks at the runs you have made so far and suggests the next few to try, where the best result is most likely to be found. Add them to the table, run them, type in the results and ask again, round after round. *Try it:* sample *Reaction yield (simulated)*.
More detail

Quality

Keeping a process stable, checking it meets its specifications, and making sure your measurements can be trusted.



Control charts

A control chart plots a measurement in time order with limits showing how far a stable process normally wanders. Points outside the limits, or unusual runs of points, are flagged for a look.

Control chart... is the standard chart for measurements: individual values, or averages and ranges of small samples. It can show different limits for each phase of a process (before and after a change), draw the limits at another number of σ, show only the latest points, or use limits saved from an earlier period. *Try it:* sample *Fill weights*. More detail

Count and rare-event chart (p, np, c, u, p', u', g, t)… charts counts instead of measurements: the share of defective items, the number of defects, or the time between rare events. For defects per area or length (u charts) the size can be a fraction, such as 2.5 m² of panel. *Try it:* sample *Daily inspection*. More detail

EWMA chart… and **CUSUM chart…** catch small, lasting shifts sooner than a standard chart, by building up evidence over several points. More detail

Multivariate control chart… watches several related measurements in one chart. *Try it:* sample *Bottle filling*. More detail

Principal-components control chart… watches many related measurements, such as a machine's sensors, through a few summary patterns learnt from a period of normal running. One chart shows rows that go unusually far along those patterns; a second shows rows that break them. *Try it:* sample *Reactor sensors*. More detail

Monitoring alert and action limits… sets alert and action levels for environmental monitoring (colony counts, positive plates or particle counts) from your own data, and charts each location against them with early warnings. *Try it:* sample *Clean-room monitoring*. More detail

Process capability

Process capability… compares the process spread with its specification limits and reports indices such as Cpk and the expected share out of specification. It can also compare groups such as machines or shifts side by side, and add a column saying whether each value is in specification. *Try it:* sample *Fill weights*. More detail

Capability summary page… puts capability, control charts and plots for one measurement on a single page. *Try it:* sample *Tablet hardness*. More detail

Between/within capability… is capability for processes made in batches, separating the variation between batches from the variation within them. *Try it:* sample *Coating batches*. More detail

Attribute capability… is capability for pass/fail or defect-count data. *Try it:* sample *Supplier lots*. More detail

Measurement systems

Before trusting a measurement, check the gauge that takes it.

Gauge R&R… splits measurement variation into repeatability (the same person measuring again) and reproducibility (different people), and compares it with the part-to-part variation. *Try it:* sample *Gauge study*. More detail

Gauge study worksheet… makes the run sheet for such a study. (Also on the Design menu.)

Bias and linearity… checks whether a gauge reads too high or too low, and whether that changes across its range. *Try it:* sample *Gauge linearity*. More detail

Gauge repeatability (Cg, Cgk)… checks a gauge by measuring one master part many times. It can also check whether readings depend on the side the gauge approaches from (hysteresis) and whether they creep over time (drift). *Try it:* sample *Gauge repeatability*. More detail

Intraclass correlation… measures how consistently different raters or instruments score the same items. More detail

Attribute agreement… checks how well inspectors agree with each other, with themselves and with the right answer when they grade items by eye. *Try it:* sample *Visual inspection*. More detail

Attribute gauge bias and repeatability… checks a go/no-go gauge. *Try it:* sample *Go/no-go gauge*. More detail

Acceptance sampling

Acceptance sampling plan… works out how many items to inspect from a lot, and how many failures to allow, to accept good lots and reject bad ones. *Try it:* sample *Supplier lots*. More detail

Problem-solving tools

Analysis of means… compares each group's average with the overall average on a chart, flagging groups that stand out. *Try it:* sample *Heat treatment*. More detail

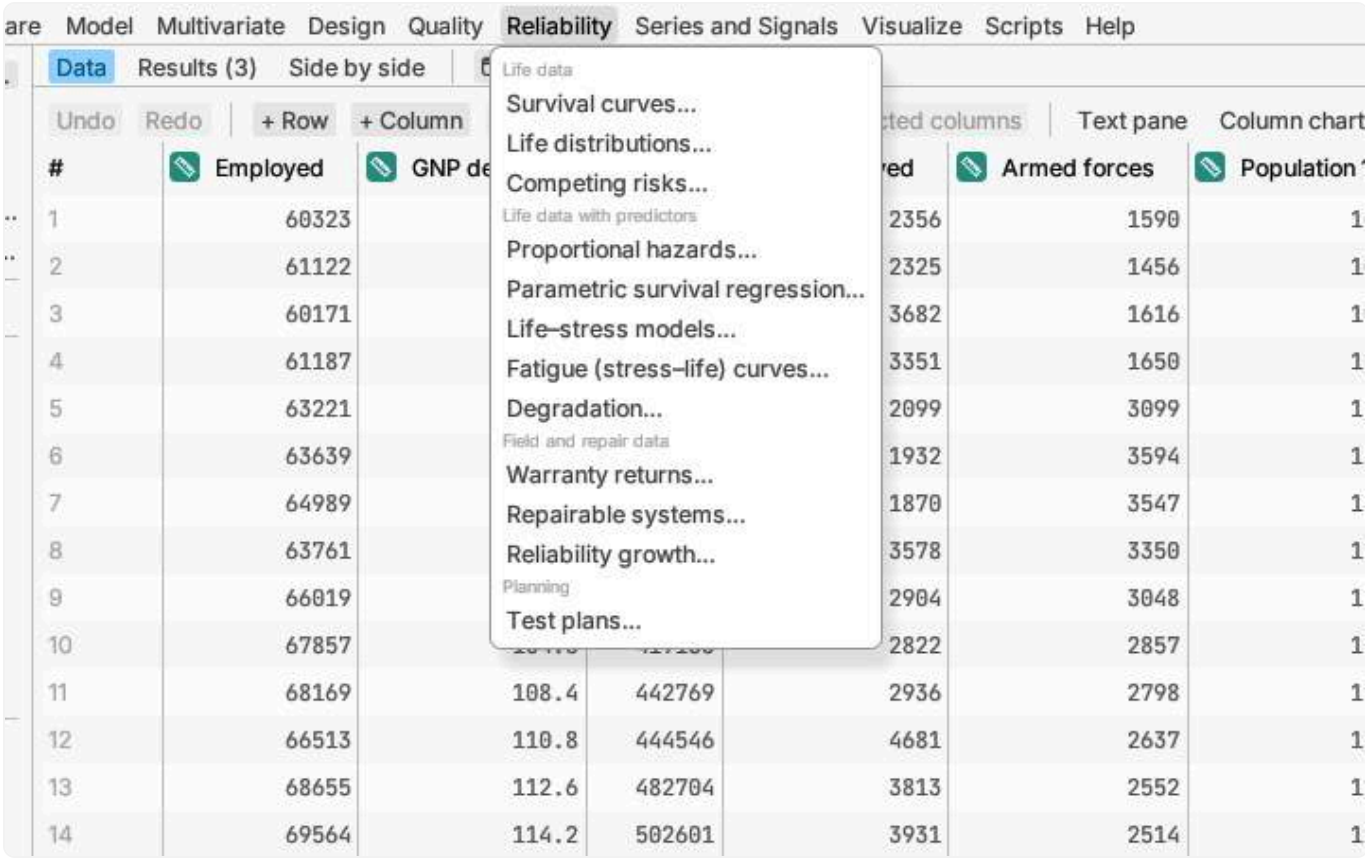
Pareto chart… ranks problem types from most to least common, so you can tackle the big few first. *Try it:* sample *Solder defects*. More detail

Cause-and-effect diagram… draws a fishbone diagram of possible causes. More detail

Multi-vari chart… shows at a glance where variation comes from: within a part, between parts or over time. *Try it:* sample *Shaft diameters*. More detail

Reliability

How long things last, and when they fail. Life data are special because some units are often still working when the study ends (*censored*): we know they lasted at least that long. These methods use those rows too.



Life data

Survival curves… draws the share of units (or patients) still surviving over time, and tests whether groups survive differently (or get steadily better or worse across ordered groups), with the number still at risk listed under the plot. *Try it:* sample *Leukemia remission*. More detail

Life distributions… fits a life distribution, such as Weibull or lognormal, to failure times, and estimates things like the time by which 10% will have failed. *Try it:* sample *Bearing life test*. More detail

Competing risks… handles units that can fail in more than one way, estimating the chance of each kind of failure. *Try it:* sample *Pump failure modes*. More detail

Life data with predictors

Proportional hazards… and **Parametric survival regression…** relate survival to other columns (treatment, age, temperature): which ones shorten or lengthen life, and by how much. *Try it:* samples *Leukemia with covariates* and *Capacitor life test*. More detail

Life–stress models… uses tests run at high stress (heat, voltage) to predict life at normal use. *Try it:* sample *Device-A accelerated life test*. More detail

Fatigue (stress–life) curves… fits the curve of cycles to failure against stress from fatigue tests, counting specimens that never broke (run-outs), and gives the stress a part can take for a given number of cycles. *Try it:* sample *Steel fatigue*. More detail

Degradation… predicts failure times from measurements that wear down over time, before anything has actually failed. *Try it:* sample *Laser degradation*. More detail

Field and repair data

Warranty returns… forecasts future returns from the sales and returns recorded so far. *Try it:* sample *Monthly warranty returns*. More detail

Repairable systems… analyses machines that are repaired and put back into service: are repairs becoming more or less frequent? *Try it:* sample *Pump repairs*. More detail

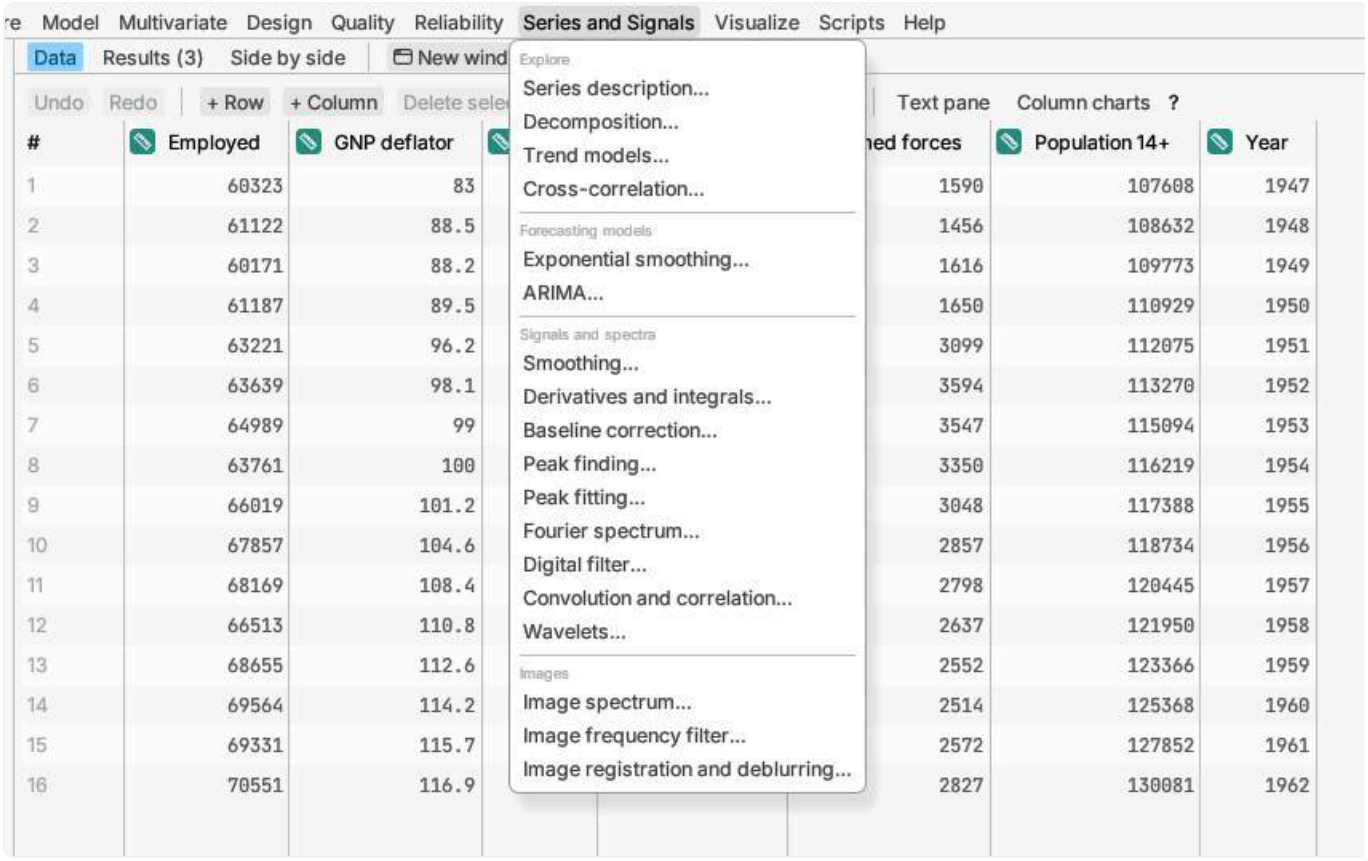
Reliability growth… tracks whether a product under development is failing less often as fixes go in. *Try it:* sample *Prototype failures*. More detail

Planning

Test plans… works out how many units to test, and for how long, to demonstrate a reliability target. It uses no data (any open table will do, even a new empty one). More detail

Series and Signals

Data recorded in order: series over time (daily sales, monthly demand), signals sampled along an axis (spectra, vibration traces) and images. Describe the pattern, forecast what comes next, filter out noise and find the frequencies hidden in the data. The rows must be in time (or x) order.



Explore

Series description... plots the series and shows how each value relates to earlier ones (autocorrelation), which reveals trends and seasonal cycles. *Try it:* sample *Monthly electricity demand*. More detail

Decomposition... splits a series into trend, seasonal pattern and what is left over. *Try it:* sample *Monthly electricity demand*. More detail

Trend models… fits a straight line or curve through time, such as an S-shaped adoption curve, and extends it. *Try it:* sample *Product adoption*. More detail

Cross-correlation… finds whether one series leads another, such as advertising spend leading sales by two weeks. *Try it:* sample *Advertising and sales*. More detail

Forecasting models

Exponential smoothing… forecasts by weighting recent values most, with trend and seasonal parts if needed. It can pick the best form for you. *Try it:* sample *Hotel occupancy*. More detail

ARIMA… is a flexible family of forecasting models built from a series' own past values and past errors; it can also search for the best model automatically. It can also use another column that drives the series, such as fuel feed driving a kiln's temperature, and forecast from planned values of it. *Try it:* samples *Reactor temperature* and *Kiln fuel feed*. More detail

Signals and spectra

Smoothing… irons out the noise in a signal measured point by point, such as a sensor trace, and draws the raw and smoothed signals together. *Try it:* sample *Machine vibration*. More detail

Derivatives and integrals… gives the rate of change and the running total (area under the curve) of y against x, and can resample the curve at evenly spaced x. *Try it:* sample *Machine vibration*, where integrating acceleration gives velocity. More detail

Baseline correction… removes the slowly varying background under a spectrum or chromatogram so the peaks sit on zero. *Try it:* sample *Raman bands*. More detail

Peak finding… lists the peaks that stand out from the noise, with their positions, heights, widths and areas. *Try it:* sample *ECG-like signal*, where the peaks are the heartbeats. More detail

Peak fitting… separates peaks that overlap by fitting a peak shape to each, gives their centres, heights, widths and areas with standard errors, and says how many peaks the data support. With many curves it fits them all and makes one table. *Try it:* sample *Raman bands*. More detail

For signals recorded many times a second (sound, vibration, sensors). Give an x (time) column or the number of samples per second, and a group column if several runs are stacked in one column.

Fourier spectrum… shows which frequencies a signal is made of, such as a 50 Hz hum, lists the strongest ones, and can draw a spectrogram of how they change over time. *Try it:* sample *Test bench signals*. More detail

Digital filter… keeps or removes a range of frequencies: smooth away fast noise, take out slow drift, or cut one interfering frequency, and save the result as a column. *Try it:* sample *Test bench signals*. More detail

Convolution and correlation… finds the delay at which two signals match best, finds repeating patterns in one signal, smooths or un-blurs a signal with a kernel column, or traces its envelope. *Try it:* sample *Test bench signals*. More detail

Wavelets… splits a signal into detail at each scale, removes noise while keeping sharp steps, and draws a scalogram of which frequencies are present at each moment. *Try it:* sample *Test bench signals*. More detail

Images

For pictures opened as data (one row per pixel, with x and y columns): microscope and camera images, surface scans, scanned prints.

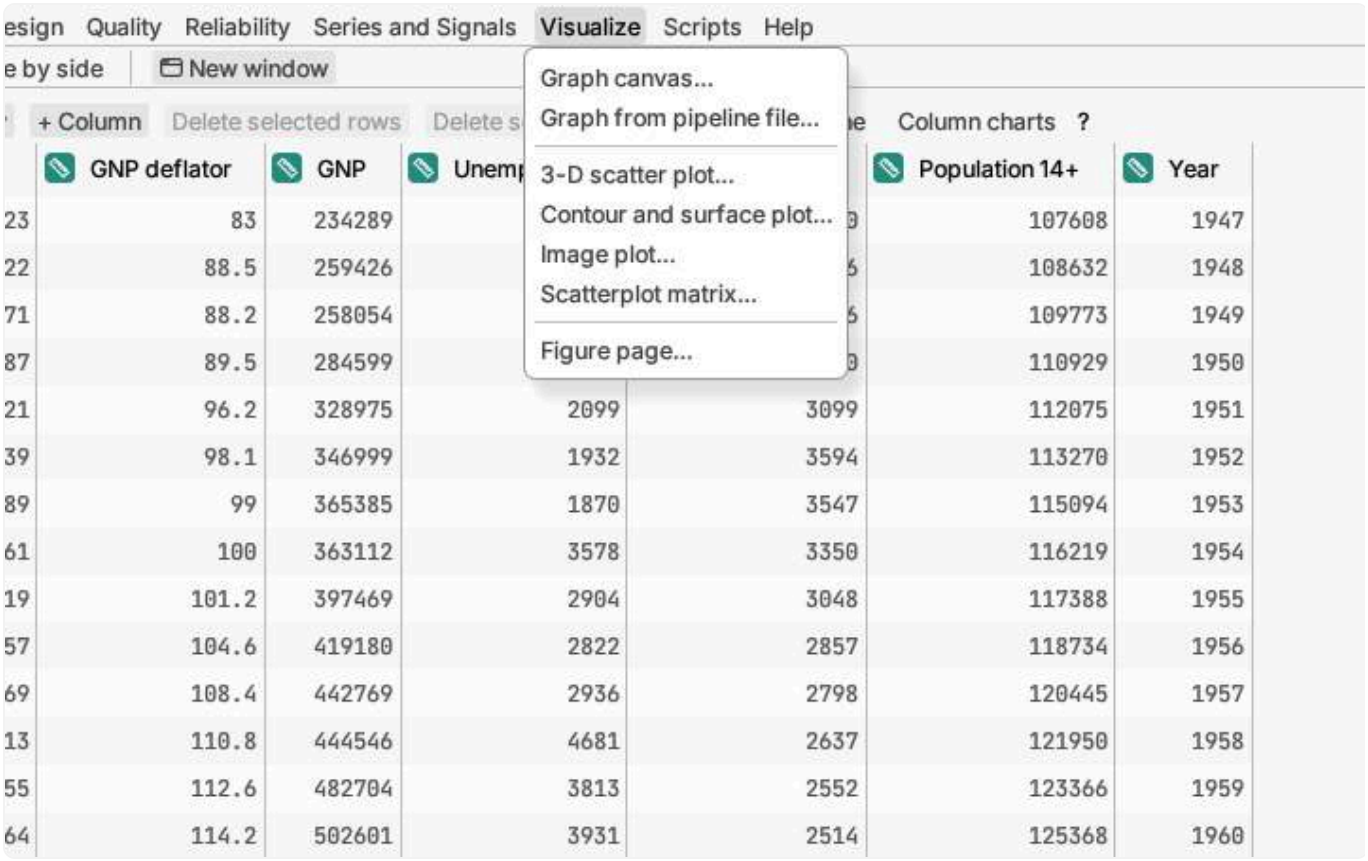
Image spectrum… shows the repeating patterns in an image as bright spots in its 2-D spectrum, lists the strongest with their spacing and direction, and shows how detail falls off from coarse to fine. Radial profiles cut the spectrum along the directions you type (across and up the image by default), so a pattern that runs one way shows as peaks in one profile and not the other. *Try it:* sample *Machined surface scan*. More detail

Image frequency filter… smooths or sharpens an image, removes slow shading, or notches out a regular pattern such as tool marks, and saves the filtered image as a column. *Try it:* sample *Machined surface scan*, where removing the tool marks reveals a pit. More detail

Image registration and deblurring… sharpens an image blurred by a known amount, finds how far a second image is shifted and lines it up, or shows the image's 2-D autocorrelation. *Try it:* sample *Blurred and shifted scene*. More detail

Visualize

Building your own graphs. (Every analysis draws its own plots too; this menu is for pictures you design yourself.)



Graph canvas… opens a blank graph. Drag columns onto the X and Y shelves, and onto colour, size or panel shelves to split the picture by group. Then add layers such as points, lines, box plots, histograms or a fitted curve. Points you select are highlighted everywhere, as in reports. You can also drop a column straight onto the graph: while you drag, the edges and centre light up (bottom for the horizontal axis, left for the vertical axis, top and right to split into panels, centre for colour). The **Row filter** can step through a column one level at a time, with a ▶ **Play** button, to watch the graph change level by level. The canvas's **Chart** list also offers pie and donut charts, radar, polar and ternary charts, wind roses, waterfall charts, ridgeline plots and vector fields; a lollipop chart is a choice in the bars layer's settings. *Try it:* samples *Harbour weather*, *Household budget* and *Soil texture*.

Put a column on the **Right axis** shelf to draw it against a second scale on the right. **Plot settings** can leave gaps in an axis (axis breaks, each piece with its own scale and share of the axis if you like) and takes titles with simple math, such as `x_1`, `x^2`, `\alpha` or

`\frac{a}{b}`. **Reference lines…** (in Plot settings, the plot's right-click menu, or at the end of the canvas's Chart row) draws lines across a plot at a value you type or at the mean, median, a percentile, mean ± some SDs or control-chart limits, which move when the data change. It also shades bands, such as a specification range, behind the data. More detail

On the canvas, the **Chart** list also offers ready-made pictures: a *clustered heat map* (a grid of colored cells with similar rows and columns put together), an *estimation plot* (how far each group's average is from a reference group, with a range of plausible values), and *Sankey*, *chord* and *network* diagrams of flows between categories. The *Brackets* layer marks which groups differ, with p-values or stars (its **…** menu picks which pairs to show, the star cut-offs and an "ns" label); the *Hulls* layer draws an outline around each group of points. *Try them:* samples *Reading trial*, *Parcel flows* and *Regional climate*.

Right-click a graph to **add text, an arrow, a line or a rectangle**; drag it to move it, double-click to edit it, or click it and nudge it with the arrow keys. **Drawing order** (at the end of the layers) puts one layer in front of another, and **Edit key…** renames, hides or reorders the entries of the key.

Graph from pipeline file… rebuilds a graph you saved earlier, on the current table, as long as it has columns of the same names. Handy for making the same chart every month.

Scatterplot matrix… draws a scatter plot for every pair of the columns you choose; it is the same command as in the Multivariate menu.

3-D scatter plot… places rows in a box by three columns; drag to turn it. *Try it:* sample *Hillside survey*. More detail

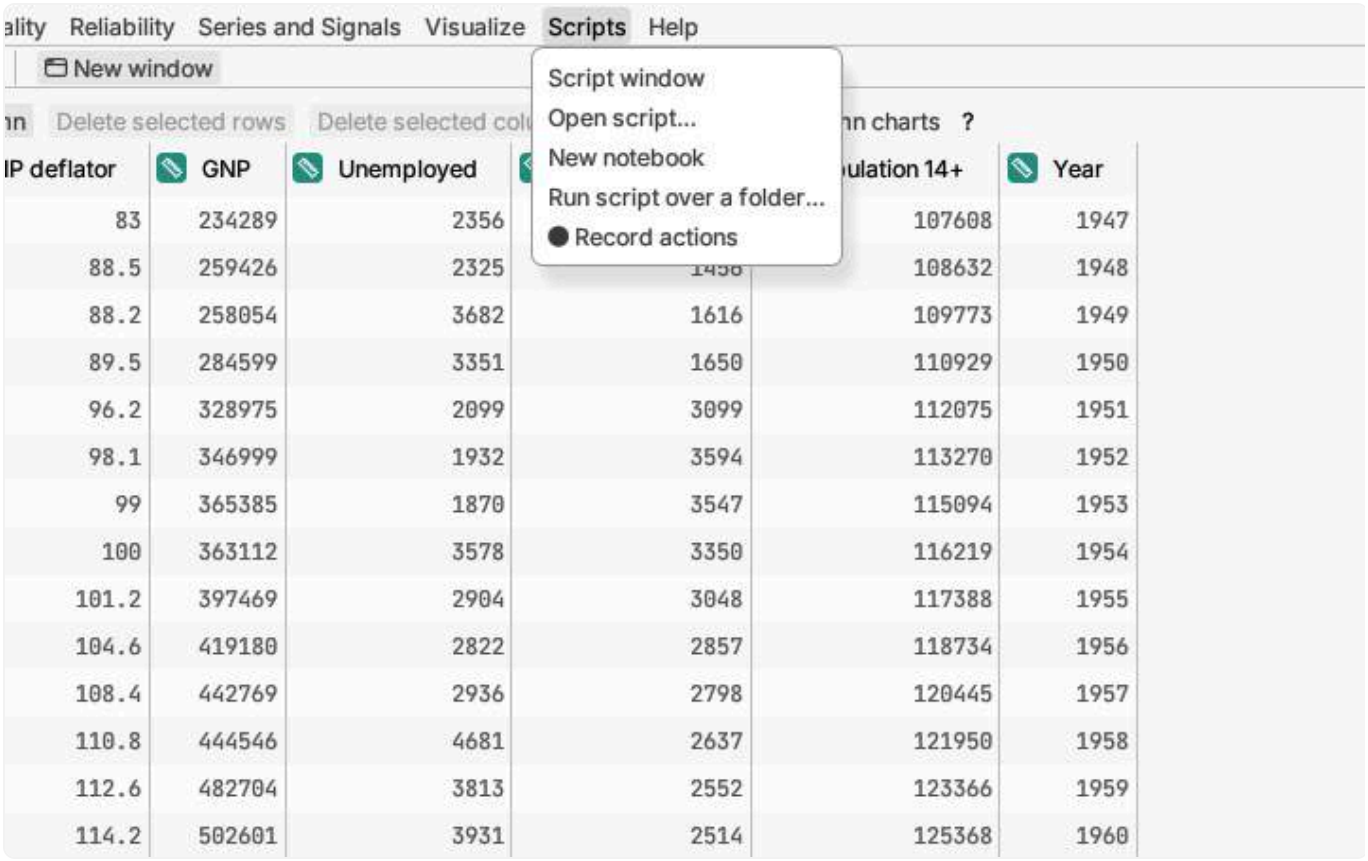
Contour and surface plot… draws a map of one column over two others, as a coloured contour map or a 3-D surface. The plot's **Plot settings** choose the colours (or plain lines). *Try it:* sample *Hillside survey*. More detail

Image plot… shows a column as a picture, one coloured square per row placed by its x and y columns: the way to look at a table opened from an image file. Drag a box over part of the picture to select those rows. More detail

Figure page… puts several plots on one page for a paper or poster, lettered A, B, C. Right-click a plot and choose **Add to figure page** to add it, then set the page size (in millimetres or inches) and the number of columns, and save the page as one picture or PDF. *Try it:* sample *Longley*, with its quantile plot and scree plot. More detail

Scripts

Repeating work automatically. Scripts are written in Rhai, a small programming language. You do not need to write one to use the app; recording is the easy way in.



Script window opens an editor where you can write and run scripts that work on your open tables.

The editor helps as you type: it suggests function and column names (press Tab to accept), highlights the matching bracket, folds long blocks away (the ⊟ marks in the margin) and finds and replaces text (**Find...**).

Open script... loads a script file into the Script window.

New notebook starts a notebook in the current table's project: a page of notes and small pieces of script, each run on its own with its results shown right under it. It is a good way to keep a worked analysis with your notes. Notebooks are saved with the project.

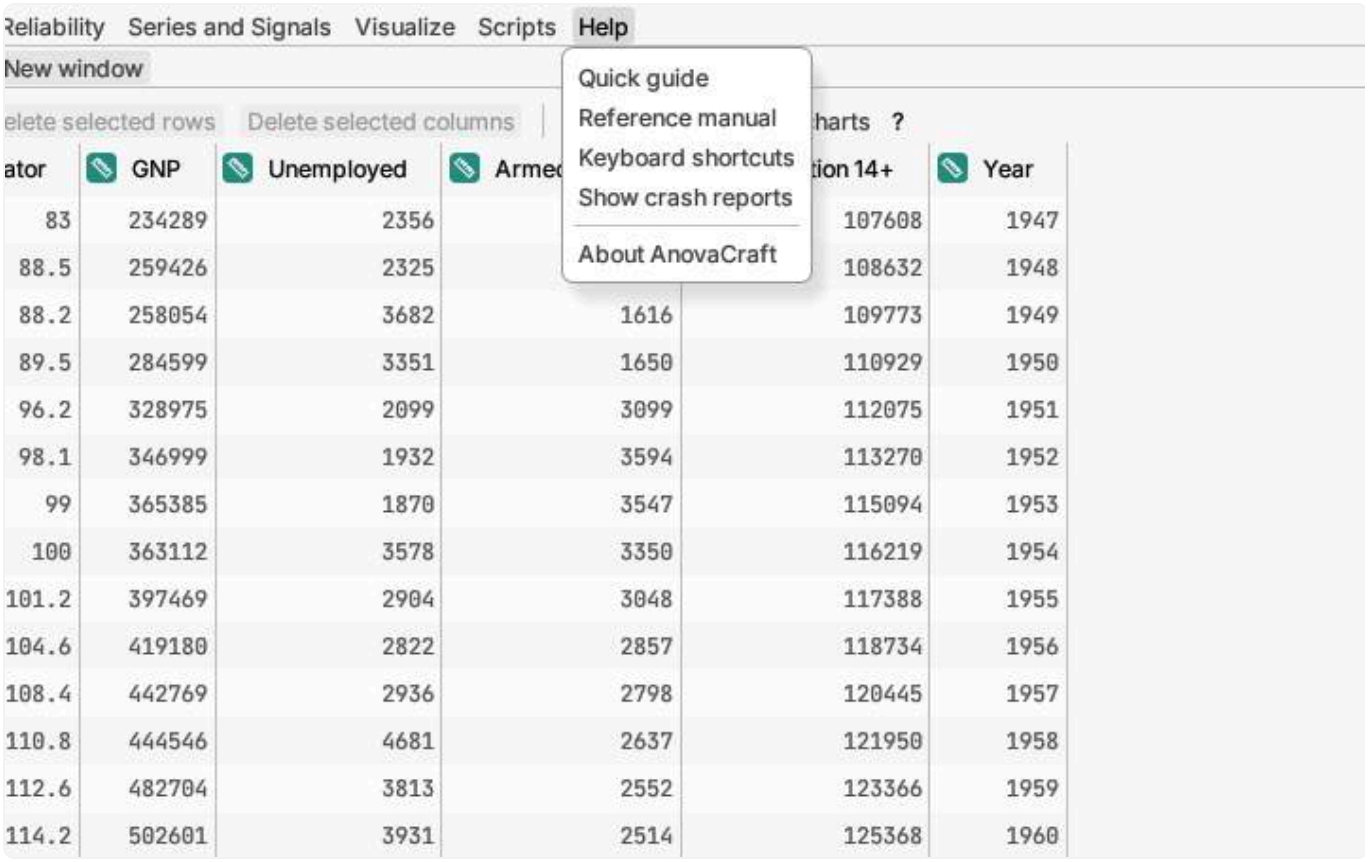
Notebooks lists the project's notebooks so you can open one again (it appears once a project has a notebook).

Run script over a folder… runs one script on every data file in a folder and collects the results in a table, for example to analyse each month's file the same way.

Record actions writes down what you do from now on (opening tables, edits, analyses and their options) as a script that repeats it. To stop, choose **Stop recording**, which takes its place in the menu; the script appears in the Script window.

More detail: Scripting and recording.

Help



Quick guide opens this guide as a PDF in your computer's PDF viewer (Preview on a Mac, your default PDF app on Windows or Linux). The guide is built into AnovaCraft, so it needs no internet connection and always matches the version you have.

Reference manual opens the Reference Manual the same way: a PDF that describes every option, report section and method in detail. Its contents page and bookmarks link to every chapter.

Keyboard shortcuts lists every shortcut key.

Show crash reports opens the folder where AnovaCraft writes a short report each time it stops on an internal error. If you report a problem, attach the newest file there.

If AnovaCraft ever closes unexpectedly, your work is not lost: every couple of minutes it keeps a copy of each project with unsaved changes, and the next time it starts it offers **Restore unsaved work**. Click **Restore** to open the copies (save them to keep the changes), **Discard** to delete them, or **Decide later** to be asked again next time.

About AnovaCraft shows the version number and AnovaCraft's licence (Apache 2.0 or MIT, whichever you prefer), with links to both manuals and to their pages online, and a note that it is an independent project. Its **Open-source licences** button lists the fonts, icons, data and software libraries AnovaCraft is built with, with each one's copyright notice and licence.

Glossary

Words you will meet in reports, in plain language. The Reference Manual gives the formulas.

Column, row. A row is one thing measured; a column is one measurement taken on each of them.

Continuous, nominal, ordinal. The kinds of column. *Continuous* columns hold measurements you can average (weight, time). *Nominal* columns hold categories with no order (supplier, colour). *Ordinal* columns hold categories with an order (small, medium, large). The icon beside each column in the sidebar shows which it is, and you can change it by right-clicking the column.

Mean, median. Two kinds of average. The *mean* adds everything up and divides by how many there are. The *median* is the middle value when they are sorted, so a few extreme values do not pull it around.

Standard deviation. How spread out the values are: roughly, the typical distance of a value from the mean.

Distribution. The overall shape of a column's values: where they cluster, how widely they spread, whether they are lopsided.

Normal distribution. The symmetric bell-shaped curve. Many methods assume data look roughly like this.

Outlier. A value far from the others. Check it before you trust or discard it.

Hypothesis test. A way of asking whether a pattern in the data could easily be due to chance alone.

p-value. How surprising the data would be if there were really no effect. Small p-values (by convention, below 0.05) suggest a real effect. The app marks them with ★. A p-value is not the size or importance of an effect.

Confidence interval. A range of plausible values for something estimated from the data, such as a mean. A 95% interval is built by a method that captures the true value 95 times out of 100.

Response, factor, predictor. The *response* is what you are trying to explain or predict. *Factors* and *predictors* are the columns you think affect it.

Model. An equation that describes how the response depends on the predictors.

R² (R-squared). The share of the response's variation that a model explains, from 0 (nothing) to 1 (everything).

Residual. The gap between an actual value and the model's prediction. Patterns in residuals hint that the model is missing something.

Interaction. When the effect of one factor depends on the level of another (for example, a fertilizer that only helps in sunny beds).

ANOVA. Analysis of variance: a family of tests that compare means by splitting the variation in the data into parts.

Design of experiments (DOE). Planning which combinations of settings to try so that a few runs tell you as much as possible.

Control limits. Lines on a control chart showing how far a stable process normally wanders. Points beyond them call for a look.

Specification limits. The limits a product must meet, set by the customer or engineer. Not the same as control limits.

Capability (Cpk). How comfortably a process fits inside its specification limits. Higher is better; 1.33 is a common target.

Censored. A unit that had not failed when observation stopped. We only know it lasted *at least* that long. Reliability methods use these rows too.

Selected, excluded, hidden. Row states. *Selected* rows are highlighted everywhere. *Excluded* rows are left out of analyses but stay in the table. *Hidden* rows are left out of plots.

Missing value, imputation. A *missing value* is an empty cell. *Imputation* fills it with an estimate; *multiple imputation* makes several filled-in copies, so results show how much the unknown values matter.

Prior, posterior. Words from Bayesian analysis. The *prior* is what you believed before seeing the data; the *posterior* is the updated belief after seeing them, given as probabilities.

Bayes factor. How many times more likely the data are under one explanation than another (for example "a difference" against "no difference"). The report labels its size: from

3 upwards (or 1/3 downwards) it calls the evidence at least moderate.